Revisiting the Global Allocation Puzzle: The Composition of Capital Flows*

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October 28, 2026

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Abstract

This paper studies the composition of cross-border capital flows by distinguishing between debt and equity assets. We show that equity inflows are positively associated with total factor productivity (TFP) growth, consistent with neoclassical theory, whereas debt inflows are negatively associated with TFP growth. The same divergence appears in the cross-section of net external positions, with fiscal capacity serving as the key country fundamental associated with these patterns. Motivated by this evidence, we develop a two-country model with heterogeneous entrepreneurs, financial constraints, and international portfolio choice to explain the joint determination of TFP and the composition of global capital flows. Differences in fiscal capacity alter both the supply of government bonds and the risk-adjusted return on domestic risky assets, producing the opposing flows of debt and equity observed in the data.

JEL codes: F21, F41, G11
Keywords: capital flows, Lucas puzzle, open-economy macroeconomics, portfolio choice

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1 Introduction

The expansion of global capital flows over the past decades has been accompanied by substantial asymmetry in external capital positions across countries. Standard neoclassical theory predicts that capital should flow from rich countries to poor countries, which exhibit lower levels of capital and higher marginal returns. However, empirical evidence often shows that fast-growing developing economies do not receive the aggregate net inflows that the theory would predict. More recently, there has been a resurgence of interest in this topic, with evidence suggesting that capital has been flowing from developing countries to developed countries (Prasad et al., 2007; Gourinchas and Jeanne, 2013).

In this paper, we document that the negative correlation between capital flows and productivity growth, known as the "allocation puzzle" by Gourinchas and Jeanne (2013), is the result of a composition effect. Using the External Wealth of Nations (EWN) database and the Penn World Table (PWT), we decompose net annual capital flows into debt and equity components from 1970 to 2023. We show that the aggregate negative relationship between capital inflows and TFP growth is driven by two opposing forces. Equity inflows are positively correlated with TFP growth, while debt inflows are negatively correlated with TFP growth. In other words, the allocation puzzle is primarily a debt flow puzzle. Equity flows move in the direction predicted by neoclassical theory, while debt flows move against it. A decomposition of the net positions, rather than flows, tells the same story. Countries with higher TFP levels have higher net equity positions and lower net debt positions. Rich countries are therefore net creditors in equity claims and net debtors in debt claims. Cross-country regressions further show that measures of fiscal and institutional capacity help explain the observed cross-country patterns in both net debt and net equity positions, while measures of capital account openness and human capital do not. This suggests that fiscal capacity is the country fundamental that is most strongly associated with the compositional patterns, which we later develop as the central feature of the model. We find that our empirical results are robust along a number of dimensions, including the sample of countries, the sample period, and more granular decompositions of asset types.

We rationalize the opposing flows of debt and equity using a two-country model with heterogeneous entrepreneurs, equity financing constraints, government bond issuance, and international portfolio choice. The framework extends the endogenous-TFP model from Buera and Moll (2015) to a novel two-country, multi-asset setting. Heterogeneous entrepreneurs draw idiosyncratic productivity and choose whether to produce or invest in financial markets. We combine this with a portfolio choice problem over home equity, foreign equity, and risk-free bonds to pin down the equilibrium net debt and equity capital flows for each country. In each country, productive entrepreneurs can scale up only by issuing risky equity subject to financing constraints, while non-producing entrepreneurs allocate their savings in an optimized portfolio. The government issues bonds backed by future tax revenue, where fiscal capacity and the expected level of entrepreneurs' profits determine the supply of safe assets.

This model generates the opposing flows of debt and equity observed in the data. The framework

is sufficiently flexible to study how a wide range of cross-country asymmetries shape the composition of capital flows (e.g., relative size of the economy, distribution of entrepreneurs' TFP, financial constraints). In this paper, we focus on differences in fiscal capacity as our leading exercise. We exogenously vary each country's ability to credibly back public liabilities with future tax revenue, which is characterized by the effective tax rate on the entrepreneurs. A country with stronger fiscal capacity can issue more government bonds, as expected tax revenue is larger. However, the higher fiscal capacity also depresses after-tax profits and lowers the risk-adjusted return on domestic equity, consequently pushing the low-productivity entrepreneurs out of production through a selection effect. As a result, investors shift away from domestic risky claims and toward bonds and foreign equity through portfolio reallocation. In equilibrium, the high-fiscal-capacity country is a net debtor in bonds and a net creditor in equity, while the low-fiscal-capacity country is a net creditor in bonds and a net debtor in equity. Because higher fiscal capacity is endogenously associated with higher observed TFP in the model through the selection effect, the model generates the opposing flows of debt and equity with respect to TFP levels that we document empirically.

Related literature and contribution. This paper is most closely related to the Lucas puzzle and the subsequent literature revisiting the puzzle (Lucas, 1990; Calvo et al., 1996). Prasad et al. (2007) document that fast-growing developing countries often export rather than import capital. Gourinchas and Jeanne (2013) sharpen this finding by showing that this "allocation puzzle" is largely a savings puzzle in which reserve accumulation by the public sector is central, complementing the public-private decomposition of Aguiar and Amador (2011) and Alfaro et al. (2014). Gourinchas and Rey (2014) document a number of stylized facts related to long-term capital flows and discuss their implications. Our results are consistent with Coppola et al. (2021), who document that bilateral portfolio investment from developed countries to large emerging markets is larger than previous estimates using a novel dataset of restated bilateral investment positions that account for cross-border financing and tax havens. Alfaro et al. (2008) provide a systematic empirical investigation and show that institutional quality is the leading explanation for why capital does not flow more strongly to poorer countries, emphasizing that capital inflows serve as a key factor linking institutional quality and long-run development.[1] Caselli and Feyrer (2007) argue that once capital is measured correctly and the differences in the price of capital goods are taken into account, the marginal product of capital is more similar across countries than previous estimates. Our paper complements this literature, but shifts the focus from the level of aggregate net flows to their composition based on the type of asset. We show that the puzzle is primarily driven by debt flows, while equity flows move in the direction predicted by neoclassical theory. This compositional fact and our model provide a bridge between the public-private distinction based on the types of asset ownership emphasized by Gourinchas and Jeanne (2013), the importance of institutional quality highlighted by Alfaro et al. (2008), and the asset type distinction that is central to our model.

[1] North (1981, 1991, 1994); Hall and Jones (1999); Acemoglu et al. (2001, 2002, 2003) discuss the role of institutions in economic development.

Our theoretical framework is most closely related to Buera and Shin (2017) and Buera and Moll (2015). Buera and Shin (2017) develop a model with heterogeneous producers, occupational choice between workers and entrepreneurs, and underdeveloped domestic financial markets to explain the joint dynamics of TFP and capital flows. Our model is closest to theirs in motivation, but adopts the occupational choice structure of Buera and Moll (2015), in which entrepreneurs choose between production and financial investment rather than between entrepreneurship and wage work. This generates a class of portfolio investors, and their asset demand determines the composition of capital flows. To our knowledge, our paper is the first to explicitly model distinct asset structures in the context of explaining the observed patterns of capital flows between rich and poor countries. We extend this framework into a multi-asset, multi-country setting where non-producing entrepreneurs solve a portfolio choice problem following the mean-variance approach of Campbell and Viceira (2002), which allows the model to speak directly to the equity-debt composition of external positions. More broadly, our paper relates to a literature that attempts to explain why capital fails to flow to fast-growing or capital-scarce economies. Gertler and Rogoff (1990) and Boyd and Smith (1997) show how frictions in local capital markets can cause capital to flow from poor to rich countries. Caballero et al. (2008) and Mendoza et al. (2009) emphasize the interaction between local and international financial markets to explain global imbalances. Angeletos and Panousi (2011) show that entrepreneurial risk combined with incomplete markets can generate global imbalances through a precautionary saving channel, and Sandri (2014) develops a related mechanism in which risky entrepreneurship generates capital outflows during growth episodes. Benhima (2013) revisits the allocation puzzle using a portfolio approach with uninsurable idiosyncratic investment risk. Matsuyama (2004) shows that financial market globalization can itself produce endogenous inequality across otherwise identical countries through a symmetry-breaking mechanism. Song et al. (2011) study China's experience, showing how the expansion of a self-financed private sector generates excess saving and capital outflows. Carroll et al. (2000) use habit formation to explain the positive correlation between saving and growth, while Aguiar and Amador (2011) provide a political economy explanation for why governments of rapidly growing countries accumulate net foreign assets.

This paper is also related to the broader literature on major puzzles in international macroeconomics. Obstfeld and Rogoff (2000) argue that many of the central puzzles, including the home-bias puzzles, the Feldstein–Horioka puzzle, and the Backus–Kehoe–Kydland puzzle, can be explained by frictions in international goods and asset markets.[2] Carroll and Weil (1994) establish a positive cross-country correlation between savings rates and growth that is difficult to reconcile with the permanent income hypothesis. Gourinchas and Jeanne (2013) show that this savings puzzle is at the heart of the allocation puzzle. Our results highlight that the relevant margin is not only how much a country saves in aggregate, but also which assets are being accumulated and issued. Aggregate savings and net flows may therefore paint an incomplete picture of the relationship between

[2]See French and Poterba (1991); McCallum (1995); Feldstein and Horioka (1980); Backus et al. (1992) for each of the puzzles.

growth, savings, and capital flows. Our empirical analysis builds on the work by Lane and Milesi-Ferretti (2007), who construct the External Wealth of Nations dataset. Obstfeld (2012) emphasizes that gross positions and their composition carry information that net flows obscure. Benigno et al. (2025) provide a complementary perspective by showing that capital flows from developing countries to the United States can reduce US productivity growth by drawing resources out of the tradable sector, highlighting that the macroeconomic consequences of capital flows depend on their nature and destination, not just their aggregate magnitude.

Finally, this paper is also related to the literature on global safe assets (Caballero et al., 2017; He et al., 2019). Caballero et al. (2008) argue that a global shortage of safe assets can explain persistent current account imbalances using a model in which exogenous differences in financial conditions shape the pattern of capital flows. Maggiori (2017) provides a micro foundation for countries' ability to supply safe assets to the rest of the world by modeling countries with varying financial development, where the most financially developed countries take on greater risk in equilibrium. Carroll and Jeanne (2009) develop a tractable model of precautionary reserve accumulation in which countries build up net foreign assets as insurance against sudden stops, providing a rationale for the large reserve holdings that our empirical analysis identifies as one of the drivers of the debt puzzle. Our paper connects the safe-asset literature to the allocation puzzle through the equity-debt decomposition. Once capital flows are disaggregated into risky and safe components, the supply of sovereign safe assets becomes a natural candidate for explaining why debt flows move in the opposite direction from equity flows. In our model, countries can differ along several dimensions, including financial conditions, but it is fiscal capacity that emerges as the key parameter generating the opposing flows. Fiscally stronger countries issue more government debt because future tax revenue can credibly support a larger supply of safe claims, attracting net debt inflows even as risky capital flows in the opposite direction.

Roadmap. The rest of the paper proceeds as follows. Section 2 presents the empirical evidence. Section 3 introduces the model in closed-economy setting, and Section 4 extends the model to a two-country setting. Section 5 illustrates the mechanism with comparative statics. Section 6 concludes.

2 The composition of global capital flows

We examine how global capital flows differ by asset class and show that the Lucas puzzle is primarily a debt phenomenon. We first describe the data and measurement choices in Section 2.1, then present the results in Section 2.2.

2.1 Data and measurement

We measure capital flows, external positions, and TFP using two standard data sources. The first is the External Wealth of Nations (EWN) database constructed by Lane and Milesi-Ferretti (2007)

and subsequently updated through 2023, which provides annual data on countries' external asset and liability positions disaggregated by instrument type. The EWN database reports positions for portfolio equity, foreign direct investment, portfolio debt, reserves, and other investment. The second data source is the Penn World Table (PWT), from which we obtain country-level total factor productivity (TFP) measures.

Our sample includes the 50 largest economies ranked by 2023 nominal GDP, as well as any Eurozone countries.[3] We then exclude major tax havens whose gross external positions are disproportionately large relative to domestic economic activity and primarily reflect the presence of multinational holding structures rather than genuine cross-border capital allocation.[4] We also exclude major oil exporters whose external positions are heavily influenced by commodity price cycles.[5] The resulting sample contains 43 countries from 1970 to 2023, yielding an unbalanced panel of approximately 2,000 observations.

We classify external positions into two broad categories, following the conceptual distinction between debt and equity claims that is central to our theoretical framework. *Equity-type* positions include portfolio equity, foreign direct investment (FDI), and other equity assets. These are claims whose returns are tied to the performance of the underlying firm or project and therefore bear risk. *Debt-type* positions include portfolio debt (government and corporate bonds), official reserves, and other debt assets. These are predominantly fixed-income claims that offer contractually specified returns and serve as safe or near-safe stores of value.[6]

We study both flows and stocks. For flows, we measure capital inflows as the change in net external positions scaled by GDP:

$$\text{Total Capital Inflow}_{i,t} = -\frac{\text{Current Account Balance}_{i,t}}{\text{GDP}_{i,t}} \times 100, \tag{1}$$

$$\text{Debt Capital Inflow}_{i,t} = -\frac{\Delta\text{Net Debt}_{i,t}}{\text{GDP}_{i,t}} \times 100, \tag{2}$$

$$\text{Equity Capital Inflow}_{i,t} = -\frac{\Delta\text{Net Equity}_{i,t}}{\text{GDP}_{i,t}} \times 100. \tag{3}$$

For stock-based analysis, we measure net external positions as a share of GDP (e.g., net equity assets to GDP and net debt assets to GDP), averaged over the sample period for each country. Note that the Lane and Milesi-Ferretti (2007) dataset accounts for valuation effects, giving a more comprehensive measure of each country's external position than official capital flows statistics (e.g., IMF IFS data).[7]

[3] This sample collectively accounts for over 90% of global GDP, and prevents countries with smaller levels of capital flows from disproportionately impacting our results.

[4] The excluded tax havens follow the definition in Coppola et al. (2021).

[5] The excluded major oil producers consist of UAE, Saudi Arabia, Norway, Qatar, Iraq, and Egypt.

[6] The classification is imperfect, as some FDI flows may involve financial structures that are more similar to debt, while some portfolio debt carries non-trivial credit risk. The broad equity-debt split nonetheless captures the economically relevant distinction between risky and safe capital flows.

[7] See Alfaro et al. (2008) for a discussion on the advantages of using the EWN dataset from Lane and Milesi-Ferretti (2007).

For the flow regressions, TFP growth is measured as the annual percentage change in the PWT's TFP-at-constant-national-prices variable, which is computed as a Solow residual after accounting for capital and labor inputs:[8]

$$\text{TFP Growth}_{i,t} = \frac{\text{TFP}_{i,t} - \text{TFP}_{i,t-1}}{\text{TFP}_{i,t-1}} \times 100. \tag{4}$$

For the stock exercises, we use the time-averaged level of TFP to relate long-run productivity differences to the composition of net external positions.

To examine the factors that may drive the observed patterns of capital flows, we obtain estimates of relevant country fundamentals from various data sources. The selection of the variables is motivated by evidence from Alfaro et al. (2008), who systematically explore explanations of the Lucas Puzzle, focusing on institutional capacity, years of schooling, and restrictions to capital mobility. We use measures of institutional capacity from Hanson and Sigman (2021), including their state capacity index, fiscal capacity index, and tax revenue from income and corporate profits as a percent of GDP. To measure human capital and capital openness, we use the human capital index from the PWT and the capital account openness measure from Chinn and Ito (2006), respectively.

2.2 Correlation between productivity growth and capital flows

Figure 1 restates the standard Lucas-puzzle finding by examining the relationship between average TFP growth and average capital inflows as a percent of GDP across countries. Aggregate capital inflows are weakly negatively related to TFP growth, suggesting that capital does not systematically flow toward faster-growing economies. This also replicates the "allocation puzzle" documented by Gourinchas and Jeanne (2013). Our sample, which extends through 2023, confirms that the aggregate pattern has persisted over recent decades.

[8] PWT normalizes TFP relative to the United States in 2017 (or, in recent vintages, 2021).

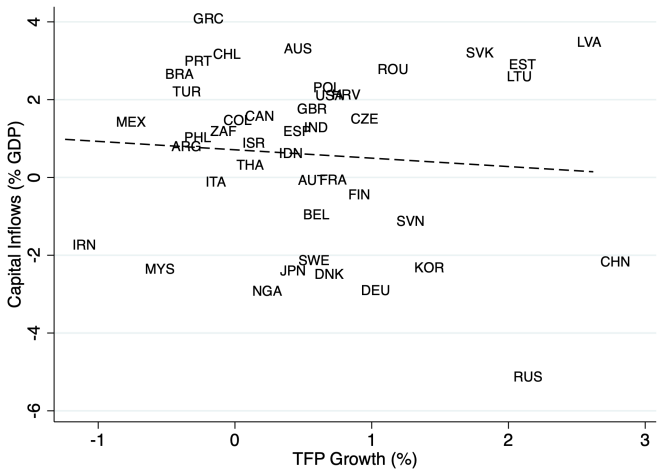
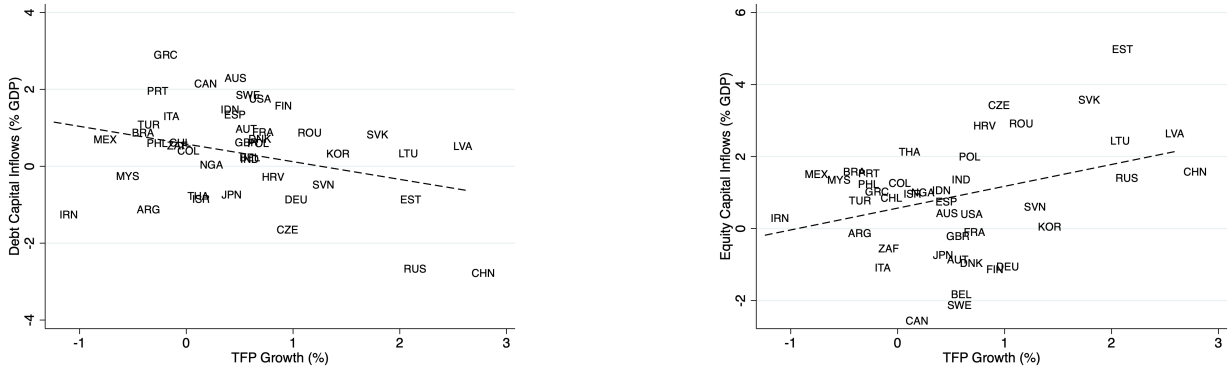


Figure 1: Aggregate capital inflows and TFP growth.

Notes: Each point is a country, showing its average levels of capital inflows (as a percent of GDP) and TFP growth between 1970 and 2023. The fitted line is from a bivariate regression of total capital inflows (as a percentage of GDP) on TFP growth.

The aggregate view, however, masks an important compositional pattern. Figure 2 decomposes capital inflows into debt and equity components. Debt inflows are negatively related to TFP growth, while equity inflows are positively related to it. The aggregate negative correlation between total inflows and TFP growth is the net result of these two opposing forces.



(a) Debt inflows

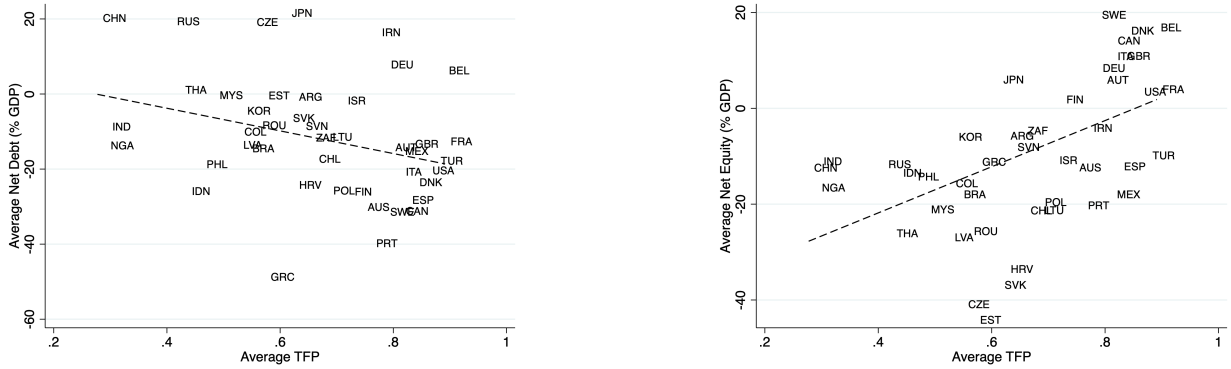
(b) Equity inflows

Figure 2: Capital inflows and TFP growth.

Notes: Panel (a) plots debt capital inflows (portfolio debt + reserves + other debt) against TFP growth. Panel (b) plots equity capital inflows (portfolio equity + FDI + other equity) against TFP growth. Each point is a country, showing its average levels of debt or equity capital inflows (as a percent of GDP) and TFP growth between 1970 and 2023.

The same pattern appears in the cross section of stock variables. Figure 3 plots time-averaged net external positions (as a share of GDP) against time-averaged TFP levels. High-TFP countries tend to hold negative net debt positions and positive net equity positions. In other words, richer and more productive economies are net issuers of debt and net creditors of equity.

These patterns are robust to alternative measurement methods of capital flows and productivity. We present the results in Appendix A.1. We find that the pattern remains the same when we substitute the global capital flow measures with the restated bilateral flows from Coppola et al. (2021), which account for the use of cross-border financing and tax havens. We also find that the results do not change when measuring TFP using the methodology from Gourinchas and Jeanne (2013).



(a) Net debt assets

(b) Net equity assets

Figure 3: NFA positions and TFP level.

Notes: Each point represents a country's time-averaged net debt (panel a) or net equity (panel b) position as a percentage of GDP, plotted against the country's time-averaged TFP level.

Cross-country analysis. To summarize the evidence more systematically and to identify which country fundamentals may help explain the observed patterns, we estimate the following cross-country regression:

$$\overline{NFA}_i = \alpha + \beta_1 \overline{\text{TFP}}_i + \beta_2 \overline{X}_i + \varepsilon_i, \tag{5}$$

where $\overline{NFA}_i$ is country i's average net foreign asset position (either net debt assets or net equity assets) as a percentage of GDP, $\overline{\text{TFP}}_i$ is the country's average TFP level, and $\overline{X}_i$ is a country fundamental of interest. All variables are averaged over the sample period. The TFP variable and any index-valued controls are standardized to mean zero and unit standard deviation.

Following Alfaro et al. (2008), we add one country fundamental at a time and examine which controls attenuate the coefficient on TFP. Their cross-country investigation of the Lucas puzzle finds that institutional quality is the leading explanation for the lack of capital flows from rich to poor countries. They also find that neither years of schooling nor restrictions on capital mobility help explain the Lucas puzzle. We perform the analogous exercise on the equity-debt decomposition of net external positions, but place additional emphasis on measures of fiscal capacity, which our model later identifies as the central margin.

Table 1 reports the results for three measures of fiscal and institutional capacity taken from Hanson and Sigman (2021): tax revenue from income and corporate profits as a percent of GDP, an

index of fiscal capacity, and an aggregate state capacity index that summarizes broader institutional capacity. All three controls attenuate the TFP coefficient for both net debt and net equity assets. For net debt assets (columns 1–4), each of the three controls renders the relationship between TFP and the net debt position statistically insignificant, and the fiscal capacity index reduces the magnitude of the TFP coefficient by the largest amount. For net equity assets (columns 5–8), none of the three controls eliminates the statistical significance of the TFP coefficient, although each attenuates it. Tax revenue from income and corporate profits has the largest attenuating effect. This motivates the focus of our model, which singles out tax and fiscal capacity as the central driver of the observed capital flow patterns.

	Net debt assets				Net equity assets			
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
TFP level (std.)	-5.090**	-4.281	-3.501	-4.514	8.121***	4.295**	7.960***	4.905*
	(2.328)	(2.890)	(2.456)	(3.713)	(1.956)	(1.926)	(2.182)	(2.525)
Tax from income/profits (% of GDP)		-0.298				1.412***		
		(0.520)				(0.331)		
Fiscal capacity (std.)			-3.614				0.368	
			(2.900)				(1.721)	
State capacity (std.)				-0.867				4.842*
				(4.103)				(2.593)
Countries	43	43	43	43	43	43	43	43

Table 1: NFA positions and TFP level: Controlling for institutional capacity

Notes: The dependent variable in each column is the corresponding net foreign asset measured as a percentage of GDP. TFP level is normalized to have a mean of zero and a unit standard deviation. Robust standard errors are reported in parentheses. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.1$.

Table 2 reports the analogous regressions with the Chinn and Ito (2006) index of capital account openness and the human capital index from the PWT. The human capital index is based on years of schooling from Barro and Lee (2013) and estimated rates of return to education. The magnitude of the TFP coefficient increases on both outcomes when either control is included. Capital account openness and human capital therefore do not appear to be the country fundamentals behind the cross-country debt and equity patterns.

	Net debt assets			Net equity assets		
	(1)	(2)	(3)	(4)	(5)	(6)
TFP level (std.)	-5.090**	-5.415*	-6.426**	8.121***	8.292***	8.691***
	(2.328)	(3.032)	(2.925)	(1.956)	(2.477)	(2.401)
Chinn-Ito index (std.)		0.662			-0.346	
		(3.176)			(2.747)	
Human capital index (std.)			3.569			-1.521
			(3.448)			(2.752)
Countries	43	43	43	43	43	43

Table 2: NFA positions and TFP level: Controlling for openness and human capital

Notes: The dependent variable in each column is the corresponding net foreign asset measured as a percentage of GDP. TFP level is normalized to have a mean of zero and a unit standard deviation. Robust standard errors are reported in parentheses. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.1$.

These cross-country regressions mirror the finding from Alfaro et al. (2008) that institutional and fiscal capacity, rather than years of education or restrictions on capital mobility, are the country fundamentals most strongly associated with the cross-country variation in net foreign positions. This directly motivates the model in Section 4, in which differences in tax and fiscal capacity across countries drive the observed patterns of external positions. In the model, a higher tax rate raises government bond supply, moving the country toward a net debtor position in bonds. It also lowers the marginal producer's after-tax return and shrinking domestic outside equity supply, which moves the country toward a net creditor position in equity.

Panel specification. We complement the cross-country evidence with a panel specification that summarizes the flow evidence and controls for unobserved heterogeneity. We estimate the following panel specification:

$$\text{Capital Inflow}_{i,t} = \alpha_i + \gamma_t + \beta\,\text{TFP Growth}_{i,t} + \varepsilon_{i,t}, \tag{6}$$

where α_i denotes country fixed effects and γ_t denotes year fixed effects. Standard errors are clustered at the country level. We use stocks in the cross-country regressions because time-averaged net positions reflect long-run accumulated holdings that map naturally to the steady-state cross-sectional patterns of interest. We use flows in the panel regressions, as year-to-year flow variation can be exploited to absorb time-invariant country-specific factors through country fixed effects.

Table 3 presents the estimates from the panel regressions. A one-percentage-point increase in TFP growth is associated with aggregate capital inflows that are 0.08 percentage points lower as a share of GDP. The estimate is statistically insignificant, but consistent in sign with the Lucas puzzle. Once we decompose the capital inflows into debt and equity, we find opposing signs. A one-percentage-point increase in TFP growth is associated with debt capital inflows that are 0.17 percentage points lower and equity capital inflows that are 0.22 percentage points higher as a share of GDP. The coefficients on both the debt and equity capital inflows are statistically significant at the 1% level. These opposing flows approximately offset, which is why the aggregate coefficient is small and insignificant.

Columns 4–7 in Table 3 further decompose the results by asset class. Within debt, we find negative relationships with TFP growth for both portfolio debt and reserves. The negative association is statistically significant for reserves. A one-percentage-point increase in TFP is associated with reserve inflows that are 0.12 percentage points lower as a share of GDP. The coefficient of -0.05 on portfolio debt is negative but statistically insignificant. The small and statistically insignificant estimate for portfolio debt may reflect the heterogeneity of this category, as it includes both government and corporate bonds that vary significantly in terms of risk. The concentration of the debt puzzle in reserves is consistent with the finding of Gourinchas and Jeanne (2013) that reserve accumulation by developing-country governments is a central driver of the allocation puzzle. Given that reserve holdings consist mainly of sovereign bonds, the higher magnitude on reserves suggests that debt flows are especially connected to sovereign or quasi-sovereign safe-asset accumulation rather

than to all debt claims uniformly. Within equity, we find that there is a positive and statistically significant correlation with TFP growth for both portfolio equity and FDI. The correlations are lower in magnitude relative to equity overall. A one-percentage-point increase in TFP growth is associated with portfolio equity and FDI inflows that are 0.14 and 0.08 percentage points higher as a share of GDP.

	(1) Aggregate	(2) Debt	(3) Equity	(4) Portf. Debt	(5) Reserves	(6) Portf. Equity	(7) FDI
TFP Growth	-0.080	-0.169***	0.216***	-0.050	-0.119***	0.143***	0.079**
	(0.058)	(0.049)	(0.062)	(0.048)	(0.029)	(0.049)	(0.032)
Observations	1816	1816	1816	1816	1816	1816	1816
Country FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes

Table 3: Asset-class decomposition of capital inflows

Notes: The dependent variable in each column is the corresponding capital inflow measured as a percentage of GDP. TFP growth is the annual percentage change in TFP from the Penn World Table. All specifications include country and year fixed effects. Standard errors, clustered at the country level, are reported in parentheses. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.1$.

The pattern documented in the panel regressions is robust along several dimensions. We present the results for all of the robustness exercises discussed below in Appendix A.3.

Given that our sample period from 1970–2023 includes several crisis episodes related to global capital flows, we estimate the baseline regression for a number of sample periods.[9] We find a consistent pattern across all of the selected sample periods. Higher TFP growth is associated with lower aggregate capital inflows, lower debt inflows, and higher equity inflows. The estimates are broadly similar in terms of magnitude, although some of the estimates with smaller samples lose statistical significance: debt inflows are significant only at the 10% level in the post-1995 sample, and equity inflows are not statistically significant in the post-2007 sample.

We also find that the results are robust to the selection of countries. When using the sample of developing countries used in Gourinchas and Jeanne (2013), the coefficient on the equity capital inflows is smaller in magnitude and statistically insignificant, suggesting that the association is weaker within developing countries relative to our baseline sample that includes many advanced economies. We also find that the results are not driven by any extreme observations. We estimate the baseline regression with a sample that excludes countries with the highest or lowest average observations of either the outcome or explanatory variable, and find that our results do not change.

Finally, rather than restricting all regressions to a common sample for which every variable is observed, we estimate each panel regression using all observations available for its outcome variable. The sample size now ranges between 1,855 and 2,071, in comparison to the sample size of 1,816 for the baseline versions. The estimated coefficients are consistent in terms of sign and magnitude,

[9]Crisis episodes during our sample period include, but are not limited to, the 1973 oil crisis, the Latin American sovereign debt crisis in the 1980s, the 1997 Asian financial crisis, the 2008 global financial crisis, and the 2020 COVID pandemic.

with statistical significance at the 1% level for the debt and equity inflows.

3 A Simple Model of Occupational Choice and Productivity

We start with a simple closed economy model to illustrate the key mechanisms. Time is discrete $t = 0, 1, 2, \ldots$, and the economy is populated by a unit mass of capitalists, a unit mass of hand-to-mouth workers, and a government.

Workers. Workers are hand-to-mouth and unconstrained in wealth.They supply one unit of labor inelastically in each period and receive government transfers T_t, which consists of the realized fiscal surplus or deficit from paying off last period's debt and the proceeds of new bond issuance. Thus, their total consumption each period equals the sum of the labor income and the transfer T_t.

Capitalists. Capitalist i enters period t with net wealth $a_{i,t}$ and draws an idiosyncratic talent $\zeta_{i,t}$, i.i.d. across capitalists and over time

$$\zeta_{i,t}, \qquad \zeta_{i,t} \sim \text{Pareto}(\eta), \qquad \Phi(\zeta) = 1 - \zeta^{-\eta}, \quad \zeta \geq 1,\ \eta > 1, \tag{7}$$

The productivity $z_{i,t}$ of a capitalist is a product of their own talent $\zeta_{i,t}$ and the country level technology A_t, i.e. $z_{i,t} = \zeta_{i,t} \cdot A_t$. For simple model, we hold the technology level A_t constant and normalize it to one. Thus, z itself is Pareto(η) and we suppress the distinction between z and ζ. Capitalists have CRRA preferences and maximize the expected discounted utility of consumption,

$$\mathbb{E}_0 \sum_{t \geq 0} \beta^t \, \frac{c_{i,t}^{1-\gamma}}{1-\gamma}, \tag{8}$$

with coefficient of relative risk aversion $\gamma > 0$. Each period, after observing his wealth and productivity draw, each capitalist chooses an occupation $o_{i,t} \in \{P, I\}$, producer or investor. A producer hires workers, issues equity, and produces output. An investor allocates his wealth between the risky equity index fund and the risk-free government bond, earning a return without participating in production.

In each period, an aggregate productivity shock $\varepsilon_t \sim \mathcal{N}(\mu, \sigma^2)$, i.i.d. over time, hits all productions after they choose their occupations. Thus, the effective productivity of capitalist i is $\tilde{z}_{i,t} = z_{i,t} \, \exp(\varepsilon_t)$.

Government. The government issues one-period risk-free bonds backed by a pledgeable fraction of expected tax revenue. Tax revenue comes from a flat tax on producers' operating profits. Since the government bond is risk-free while realized tax revenue is stochastic due to aggregate shocks, the ex-post surplus or deficit is rebated to the hand-to-mouth workers as the lump sum transfer and the proceeds of bond issuance are likewise transferred to workers.

To summarize, the timing within each period is as follows. A capitalist enters with wealth $a_{i,t}$ and observes his productivity $z_{i,t}$. He then chooses his occupation $o_{i,t}$ by comparing the return he could achieve as a producer against the return as an investor. In other words, the optimal scale and equity issuance $E_{i,t}$, or the optimal portfolio share θ_t, are solved jointly with the occupational choice. The government issues its bonds B_t over the same period. The aggregate productivity shock ε_t realize after these choices are made; the labor market then clears at the wage w_t, and output, taxes, and asset returns settle. Each capitalist ends the period with cash on hand equal to his wealth scaled by the realized return on his chosen occupation, $m_{i,t} = \mathcal{R}^o_{i,t}\, a_{i,t}$, which he divides between consumption and next-period wealth, $c_{i,t} + a_{i,t+1} = m_{i,t}$. We describe the detailed production return in Section 3.1, the asset market in Section 3.2 and the investment and occupational choice in Section 3.3. Unless stated otherwise, $\mathbb{E}$ denotes the expectation over ε_t. We drop the subscripts i and t when no confusion arises; the fiscal parameters (τ, φ) and the financing capacity ξ are constant.

3.1 Production and cost of capital

A *producer* operates the Cobb-Douglas technology

$$y_{i,t} = (\tilde{z}_{i,t}\, k_{i,t})^{\alpha} l_{i,t}^{1-\alpha} \tag{9}$$

He hires workers $l_{i,t}$ at the national wage w_t and raises capital $k_{i,t}$. Unlike standard models in which producers rent capital through debt contracts, here a producer scales production beyond his own wealth by issuing outside equity $E_{i,t}$, subject to the financing constraint $E_{i,t} \leq \xi a_{i,t}$. Each unit of operating capital trades at the market price $q(z_{i,t})$, specified below. This price rises with the producer's productivity $z_{i,t}$: capital operated by a more productive producer is worth more. Issuing equity transfers a fraction $1 - s(z_{i,t})$ of firm ownership to outside holders, who then claim $1 - s(z_{i,t})$ of firm value. Given the market price $q(z_{i,t})$, the firm is worth $V_{i,t} = q(z_{i,t})\, k_{i,t}$, and the amount of equity raised follows from firm value and the share offered:

$$E_{i,t} = \big(1 - s(z_{i,t})\big)\, q(z_{i,t})\, k_{i,t} \tag{10}$$

and thus the total operating capital is $k_{i,t} = a_{i,t} + E_{i,t}$. Given the equity price $q(z_{i,t})$ and the state-contingent wage $w_t(\varepsilon_t)$, the producer commits operating capital $k_{i,t}$ before the shock and hires labor $l_{i,t}$ after it, to maximize after-tax profits

$$\begin{aligned} \max_{k_{i,t},\, l_{i,t}} \; & s(z_{i,t}) \left[(1-\tau) \left[(\tilde{z}_{i,t}\, k_{i,t})^{\alpha} l_{i,t}^{1-\alpha} - w_t(\varepsilon_t)\, l_{i,t} \right] + (1-\delta) k_{i,t} \right] \\ \text{s.t.} \quad & E_{i,t} = \big(1 - s(z_{i,t})\big)\, q(z_{i,t})\, k_{i,t}, \\ & k_{i,t} = a_{i,t} + E_{i,t}, \\ & E_{i,t} \leq \xi a_{i,t}. \end{aligned}$$

The two inputs are committed at different dates, and that difference is the source of the producer's risk. Capital is chosen before the shock: the producer issues equity at the pre-shock price $q(z_{i,t})$ and installs $k_{i,t} = a_{i,t} + E_{i,t}$, a position he cannot revise once ε_t is realized. Labor is hired after the shock, on a spot market that clears at the realized wage $w_t(\varepsilon_t)$, and its bill is paid out of contemporaneous revenue, so labor requires no financing and no advance commitment. The producer therefore bears aggregate risk on his installed capital while choosing labor freely in the realized state. This timing lets us solve the problem backward. Given installed capital $k_{i,t}$ and the realized shock, optimal labor demand $l^*_{i,t}$ follows from the static first order condition; the pre-shock capital choice then takes into account the resulting stochastic profit. The labor first order condition gives

$$l^*_{i,t}(\varepsilon_t) = \left(\frac{1-\alpha}{w_t(\varepsilon_t)}\right)^{1/\alpha} \tilde{z}_{i,t}\, k_{i,t}, \qquad \Pi_{i,t} = \pi_t(\varepsilon_t)\, \tilde{z}_{i,t}\, k_{i,t}, \qquad \pi_t(\varepsilon_t) \equiv \alpha \left(\frac{1-\alpha}{w_t(\varepsilon_t)}\right)^{(1-\alpha)/\alpha} \tag{11}$$

And the payoff per unit of operating capital is

$$\begin{aligned} \mathcal{R}^K(z_{i,t}, \varepsilon_t) &= (1-\tau)\, \pi_t(\varepsilon_t)\, z_{i,t}\, \exp(\varepsilon_t) + (1-\delta) = (1-\tau)\, z_{i,t}\, \mathcal{D}_t(\varepsilon_t) + (1-\delta), \\ \mathcal{D}_t(\varepsilon_t) &\equiv \pi_t(\varepsilon_t)\, \exp(\varepsilon_t), \end{aligned} \tag{12}$$

where $\mathcal{D}_t(\varepsilon_t)$ is stochastic through two channels: the direct shock $\exp(\varepsilon_t)$ and the market clearing wage $w_t(\varepsilon_t)$. Appendix B.1 derives the wage from labor-market clearing, $w_t(\varepsilon_t) = (1-\alpha)\big(\exp(\varepsilon_t)\tilde{K}_t\big)^{\alpha}$, where $\tilde{K}_t$ is the productivity-weighted capital of the active producers. By the envelope theorem, $\pi_t(\varepsilon_t)\exp(\varepsilon_t) z_{i,t}$ is the marginal product of capital once labor is hired optimally.

Outside equity is a claim on the firm's payoff, so equity price is what the marginal buyer, the investor, assigns to that payoff. Let $\mathcal{M}$ denote the investor's stochastic discount factor, derived below in the investor's problem. Then the price per unit of operating capital is

$$q(z_{i,t}) = \mathbb{E}\big[\mathcal{M}\, \mathcal{R}^K(z_{i,t}, \varepsilon_t)\big]. \tag{13}$$

and inherits the affine form

$$q(z_{i,t}) = (1-\tau)\, z_{i,t}\, p_t + \frac{1-\delta}{r_t}, \qquad p_t \equiv \mathbb{E}[\mathcal{M}\mathcal{D}_t], \qquad r_t \equiv \frac{1}{\mathbb{E}[\mathcal{M}]}. \tag{14}$$

Substituting $k_{i,t} = a_{i,t} + E_{i,t}$, the producer's net return on his own wealth is

$$\mathcal{R}^P(z_{i,t}, \varepsilon_t) = \frac{s(z_{i,t})\, \mathcal{R}^K(z_{i,t}, \varepsilon_t)\, k_{i,t}}{a_{i,t}} = \left[1 + \frac{E_{i,t}}{a_{i,t}}\left(1 - \frac{1}{q(z_{i,t})}\right)\right] \mathcal{R}^K(z_{i,t}, \varepsilon_t). \tag{15}$$

The producer's issuance decision depends only on the coefficient $1 + \frac{E_{i,t}}{a_{i,t}}\left(1 - \frac{1}{q(z_{i,t})}\right)$. The per-unit gain from issuing equity, $1 - \frac{1}{q(z_{i,t})}$, is deterministic: issuing raises the return if and only if the market values a unit of installed capital above its replacement cost. The optimal choice is therefore a corner solution. The producer self-finances, $E_{i,t} = 0$, when $q(z_{i,t}) < 1$, and issues to the cap, $E_{i,t} = \xi a_{i,t}$, when $q(z_{i,t}) \geq 1$. Every active producer has $q(z_{i,t}) \geq 1$ in equilibrium (Proposition 3.4),

so the return to producing is

$$\mathcal{R}^P(z_{i,t},\varepsilon_t) = \left[1+\xi\left(1-\frac{1}{q(z_{i,t})}\right)\right]\mathcal{R}^K(z_{i,t},\varepsilon_t). \tag{16}$$

This return combines the return on own capital, $\mathcal{R}^K(z_{i,t},\varepsilon_t)$, with the gain from equity financing, $\xi\left(1-\frac{1}{q(z_{i,t})}\right)\mathcal{R}^K(z_{i,t},\varepsilon_t)$. Each unit of equity raises capital that earns $\mathcal{R}^K$ at a cost of equity $\mathcal{R}^K/q(z_{i,t})$, generating a net gain of $\mathcal{R}^K(1-1/q(z_{i,t}))$. Since $q(z_{i,t})$ rises with productivity, more productive producers both earn higher returns and face a lower cost of equity, so the gains from equity issuance increase with $z_{i,t}$.

3.2 Equity index fund and government bonds

There are two financial assets traded in the economy: a risky equity index and a safe bond. The equity index is a claim to the pool of heterogeneous equities issued by all producers, and the bond is issued by the government that is risk-free and backed by its expected tax revenue. Let $\mathcal{P}_t \equiv \{i : o_{i,t} = P\}$ denote the endogenous set of capitalists who choose to produce. The index payoff and its price are

$$\Pi^e_t(\varepsilon_t) = \int_{i\in\mathcal{P}_t}\big(1-s(z_{i,t})\big)\,k_{i,t}\,\mathcal{R}^K(z_{i,t},\varepsilon_t)\,di, \qquad P^e_t = \mathbb{E}[\mathcal{M}\,\Pi^e_t] = \int_{i\in\mathcal{P}_t}\big(1-s(z_{i,t})\big)\,k_{i,t}\,q(z_{i,t})\,di, \tag{17}$$

Proposition 3.1 (Equity index return)**.** *The return on the equity index is the average cost of equity of active producers, and it is priced by the investor's stochastic discount factor:*

$$\mathcal{R}^e_t(\varepsilon_t) = \frac{\Pi^e_t(\varepsilon_t)}{P^e_t} = \mathbb{E}_z\left[\frac{\mathcal{R}^K(z_{i,t},\varepsilon_t)}{q(z_{i,t})}\;\middle|\; i\in\mathcal{P}_t\right], \qquad \mathbb{E}_\varepsilon\big[\mathcal{M}\,\mathcal{R}^e_t(\varepsilon_t)\big] = 1, \tag{18}$$

where $\mathbb{E}_z$ averages over the cross section of producers' productivitiy distribution and $\mathbb{E}_\varepsilon$ over the aggregate shock.

The proof is in Appendix B.2. Each active producer's index weight is the market value of his outside equity, which equals $\xi a_{i,t}$ under the binding financing constraint and independent of his productivity. The value-weighted index return therefore collapses to the simple cross sectional average of the producers' cost of equity $\mathcal{R}^K(z_{i,t},\varepsilon_t)/q(z_{i,t})$. Since every firm loads on the common aggregate risk, the index remains risky.

The safe asset in this economy is the government bond. The government issues a one-period bond following the fiscal rule

$$r_t B_t = \varphi\,\mathbb{E}[\tau\Pi^A_t], \qquad \Pi^A_t = \mathcal{D}_t\tilde{K}_t, \qquad \tilde{K}_t \equiv \int_{i\in\mathcal{P}_t} z_{i,t}\,k_{i,t}\,di, \tag{19}$$

where Π^A_t denotes aggregate operating profit, $\tilde{K}_t$ the productivity-weighted operating capital of active producers, and φ the pledgeable fraction of expected tax revenue. Since workers absorb the

ex-post gap between the committed repayment $r_t B_t$ and realized revenue $\tau \Pi_t^A$, the bond pays the gross return r_t in every state. The fiscal parameters (τ, φ) govern the government's capacity to manufacture safe assets out of risky output: τ is the distortionary tax and φ the non-distortionary fiscal capacity of a government. Safe-asset supply is thus set by fiscal policy, and the risk-free rate is pinned down by bond market clearing.

3.3 Investment and occupational choice

An investor allocates his wealth between the risk-free government bond and the risky equity index. The financial portfolio return is

$$\mathcal{R}_t^I(\varepsilon_t) = r_t + \theta_t\big(\mathcal{R}_t^e(\varepsilon_t) - r_t\big), \tag{20}$$

where θ_t is the portfolio share in the equity index. The optimal share, and the stochastic discount factor that it implies, follow from the capitalist's dynamic problem.

The aggregate state of the economy is capitalists' aggregate wealth $W_t \equiv \int a_{i,t}\, di$. Since the idiosyncratic productivity level is redrawn every period, the model features an aggregate state variable W that do not require tracking the underlying distriubtion of (a_i, z_i). In other words, the wage, the risk-free rate, and the price of risk depend on W_t through market clearing, and thus the return to either occupation as well as the next period's aggregate wealth level depend on the current aggregate wealth W. Let M_t denote aggregate capitalist cash on hand after the shock and $W_{t+1} = \Gamma(W_t, \varepsilon_t)$ the law of motion of aggregate wealth. Because W_{t+1} depends on the shock, the value of cash on hand at the end of the period is a random variable when occupations and portfolios are chosen. This stochastic end of period cash on hand results in the stochastic weight that charaterise the certainty equivalents and the stochastic discount factor below.

Consider a capitalist who enters the period with wealth a and productivity z when aggregate wealth is W. Following the timing of the model, he chooses an occupation and the corresponding production or portfolio decision x before the shock, and consumes and saves out of the realized cash on hand $m^o = \mathcal{R}^o a$ after it. His value function is

$$\begin{aligned} V(a, z; W) &= \max_{o \in \{P, I\}} \; \max_{x} \; \mathbb{E}_\varepsilon\Big[U\big(\mathcal{R}^o(z, \varepsilon; x)\, a;\, \Gamma(W, \varepsilon)\big)\Big], \\ U(m; W') &= \max_{0 \le a' \le m} \Big\{u(m - a') + \beta\, \mathbb{E}_{z'}\big[V(a', z'; W')\big]\Big\}, \end{aligned} \tag{21}$$

where $u(c) = c^{1-\gamma}/(1-\gamma)$, $\mathbb{E}_\varepsilon$ is the expectation over the shock, $\mathbb{E}_{z'}$ the expectation over next period's draw, and $U(m; W')$ is the value of end-of-period cash on hand m when next period's aggregate wealth is W'. The returns $\mathcal{R}^P(z, \varepsilon; x)$ and $\mathcal{R}^I(\varepsilon; \theta)$ are those of the previous subsections, evaluated at the prices of the state W.

The value of cash on hand depends on $W' = \Gamma(W, \varepsilon)$, which is stochastic when the occupation is chosen. For a random variable Λ and a random variable Y, denote $\mathbb{E}^\Lambda[Y] \equiv \mathbb{E}[\Lambda Y]/\mathbb{E}[\Lambda]$ for the expectation under the shock distribution tilted by Λ, and for a gross return $\mathcal{R}$ define the weighted

certainty equivalent

$$CE^{\Lambda}(\mathcal{R}) \equiv \left(\mathbb{E}^{\Lambda}[\mathcal{R}^{1-\gamma}]\right)^{1/(1-\gamma)}. \tag{22}$$

With log utility, it is the certainty equivalent $CE(\mathcal{R}) \equiv (\mathbb{E}[\mathcal{R}^{1-\gamma}])^{1/(1-\gamma)}$ of a one-period problem.

Proposition 3.2 (Occupational choice rule)**.** *Let $\gamma > 0$, $\gamma \neq 1$, and suppose that the continuation coefficient $\bar{\psi}(W')$ defined in Appendix B.3 is finite at every W'.*

(a) The value of end-of-period cash on hand is $U(m; W') = \Lambda(W')\, m^{1-\gamma}/(1-\gamma)$ with $\Lambda(W') = \left(1-\varsigma(W')\right)^{-\gamma}$, where $\varsigma(W')$ is the fraction of cash on hand that every capitalist saves. When occupations and portfolios are chosen, $W' = \Gamma(W, \varepsilon)$ depends on the shock, so $\Lambda_t \equiv \Lambda(W_{t+1})$ is a random variable, common to all capitalists. The value of occupation o is

$$V^o(a, z; W) = \mathbb{E}[\Lambda_t]\, \frac{CE^{\Lambda}\left(\mathcal{R}^o(z)\right)^{1-\gamma}}{1-\gamma}\, a^{1-\gamma}, \tag{23}$$

where $\mathcal{R}^o(z)$ embeds the financial choice that maximizes $CE^{\Lambda}\left(\mathcal{R}^o(z; x)\right)$ over x, and the capitalist produces if and only if $CE^{\Lambda}\left(\mathcal{R}^P(z, \varepsilon)\right) \geq CE^{\Lambda}\left(\mathcal{R}^I(\varepsilon)\right)$.

(b) The occupational choice is independent of wealth and carries no option value. There is a cutoff $\bar{z}$ such that capitalists with $z \geq \bar{z}$ produce and those with $z < \bar{z}$ invest, and the marginal capitalist is indifferent between the two occupations:

$$CE^{\Lambda}\left(\mathcal{R}^P(\bar{z}, \varepsilon)\right) = CE^{\Lambda}\left(\mathcal{R}^I(\varepsilon)\right). \tag{24}$$

(c) With log utility, $U(m; W') = \ln m/(1-\beta) + \ell(W')$ for a function ℓ of the state, every capitalist saves the fraction β, the weights drop out, and the capitalist produces if and only if $\mathbb{E}[\ln \mathcal{R}^P(z, \varepsilon)] \geq \mathbb{E}[\ln \mathcal{R}^I(\varepsilon)]$.

The proof is in Appendix B.3. The occpuational choice has no option value because the idiosyncratic producion level is redrawn every period and next period's aggregate state is the same whichever occupation the capitalist chooses today. The weight Λ_t reflects the return distribution in the next aggregate state, and thus affect the occupational choice as the capitalists also care about the covariance between his return and the aggreagte state. Returns are therefore valued under the Λ_t-tilted shock distribution rather than the physical distribution. Because producers and investors face the same aggregate weight Λ_t , the cutoff structure is preserved.

Corollary 3.1 (Consumption policy)**.** *Every capitalist, regardless of occupation, saves the common fraction $\varsigma(W_{t+1})$ of end-of-period cash on hand and consumes the rest:*

$$a_{t+1} = \varsigma(W_{t+1})\, m_t, \qquad c_t = \left(1-\varsigma(W_{t+1})\right) m_t, \qquad m_t = \mathcal{R}^o_t\, a_t. \tag{25}$$

The saving rate solves $\varsigma/(1-\varsigma) = \big(\beta\,\bar{\psi}(W_{t+1})\big)^{1/\gamma}$*, where* $\bar{\psi}(W_{t+1})$ *is the expected coefficient* $\mathbb{E}[\Lambda_t]$ *of the value function (Appendix B.3), and it is known once the shock is realized and is the same for every capitalist.*

The proof is in Appendix B.3.

The investor's portfolio and the stochastic discount factor. An investor chooses the equity share θ that maximizes $CE^{\Lambda}(\mathcal{R}^I)$, which for $\gamma \neq 1$ is the same as maximizing $\mathbb{E}[\Lambda_t(\mathcal{R}^I)^{1-\gamma}]/(1-\gamma)$. The objective is the same for every investor and strictly concave in θ, so the optimal share is unique and common to all investors, and it solves the first-order condition

$$\mathbb{E}\big[\Lambda_t\,(\mathcal{R}^I)^{-\gamma}\big(\mathcal{R}^e(\varepsilon) - r\big)\big] = 0. \tag{26}$$

Because investors are the marginal holders of both assets, the stochastic discount factor $\mathcal{M}$ that prices the equity index and the government bond follows from the investor's first-order condition. The next proposition characterizes the stochastic discount factor and its implications for the cost of capital.

Proposition 3.3 (The stochastic discount factor)**.** *In equilibrium, the investor's stochastic discount factor that prices all assets in the economy is*

$$\mathcal{M} = \frac{\Lambda_t\,(\mathcal{R}^I)^{-\gamma}}{\mathbb{E}[\Lambda_t\,(\mathcal{R}^I)^{1-\gamma}]}, \tag{27}$$

with Λ_t *the continuation weight of Proposition 3.2. Moreover, the risk-free rate* r*, the equity index return* $\mathcal{R}^e$*, and the equity price* $q(z)$ *satisfy*

$$r = \frac{1}{\mathbb{E}[\mathcal{M}]}, \qquad p = \mathbb{E}[\mathcal{M}\mathcal{D}] \qquad q(z) = \mathbb{E}\big[\mathcal{M}\,\mathcal{R}^K(z,\varepsilon)\big] = (1-\tau)\,z\,p + \frac{1-\delta}{r}. \tag{28}$$

The proof is in Appendix B.4. The price of the risky component of production, $p = \mathbb{E}[\mathcal{M}\mathcal{D}]$, decomposes as

$$p = \frac{\mathbb{E}[\mathcal{D}]}{r} + \mathrm{Cov}(\mathcal{M}, \mathcal{D}). \tag{29}$$

3.4 Aggregation and equilibrium conditions

The occupational cutoff divides the population of capitalists into producers and investors. The investor's return is the producer's outside option, so $CE^{\Lambda}(\mathcal{R}^I)$ is the *hurdle rate* of the economy: a capitalist produces only if the risk-adjusted return on running his firm exceeds the return he could earn holding financial assets. Holding the price of risk p and the portfolio composition θ fixed, a higher risk-free rate raises the return on the outside option and lowers $q(z)$, tightening selection; a higher price of risk raises the value of the profit claim and the gains from issuing equity, loosening selection. The hurdle rate also implies that in eqilibrium producers will choose to issue equity until

the financing constraint binds. Let z_q denote the issuance threshold at which the equity price of capital reaches par, $q(z_q) = 1$.

Proposition 3.4 (Maximum issuance of equity)**.** *In equilibrium, the occupational cutoff lies weakly above the issuance threshold, $\bar{z} \geq z_q$ and thus every producer has his equity price $q(z) \geq 1$ and issues to the financing cap, $E = \xi a$.*

A capitalist whose capital is worth less than replacement cost always earns more by holding financial assets than by producing himself. So every producer faces an equity price of capital of at least one, and issues to the financing cap. The proof, in Appendix B.5, formalizes this argument. The same conclusion holds in the two-country model of Section 4.

Let $W \equiv \int a_i \, di$ denote aggregate capitalist wealth. The mass of investors is $\Phi(\bar{z}) = 1 - \bar{z}^{-\eta}$, and the mass of producers is $1 - \Phi(\bar{z}) = \bar{z}^{-\eta}$. Thus

$$W^I = \Phi(\bar{z})\, W = (1 - \bar{z}^{-\eta})\, W, \qquad W^P = \bar{z}^{-\eta}\, W. \tag{30}$$

so the capital stock and its productivity-weighted counterpart are

$$K = (1 + \xi)\, W^P, \qquad \tilde{K} = \text{TFP} \times K = (1 + \xi)\, \frac{\eta}{\eta - 1}\, \bar{z}^{1-\eta}\, W. \tag{31}$$

The following proposition shows that the aggregate production function takes the Cobb–Douglas form, with the country level aggergate TFP determined by the aggregate shock and the occupational cutoff.

Proposition 3.5 (Aggregate production function)**.** *In equilibrium, aggregate output takes the Cobb–Douglas form*

$$Y = Z\, K^{\alpha} L^{1-\alpha}, \qquad Z = \exp(\alpha\varepsilon)\big(\mathbb{E}[z \mid z \geq \bar{z}]\big)^{\alpha} = \exp(\alpha\varepsilon)\, \text{TFP}^{\alpha}, \tag{32}$$

With unit aggregate labor supply $L = 1$, we have $Y = \exp(\alpha\varepsilon)\, \tilde{K}^{\alpha} = \exp(\alpha\varepsilon)\, \big(\text{TFP} \times K\big)^{\alpha}$. And the labor income and aggregate producer pre-tax profits are

$$w\, L = (1 - \alpha)\, Y, \qquad \Pi^A = \alpha\, Y. \tag{33}$$

Appendix B.6 shows the proof of the proposition. Static aggregate TFP contains both the aggregate shock and the selection margin. The cost of capital reaches aggregate productivity through the cutoff $\bar{z}$, which moves output along two opposing margins: a higher $\bar{z}$ raises average productivity, as TFP $\propto \bar{z}$, but reduces the mass of producers and operating capital, $\tilde{K} \propto \bar{z}^{1-\eta}$. Output is absorbed by consumption and investment:

$$Y_t = \underbrace{C_t^W + C_t^P + C_t^I}_{\text{consumption}} + I_t, \qquad I_t = K_{t+1} - (1 - \delta) K_t. \tag{34}$$

Producers and investors each consume the fraction $1-\varsigma_t$ of their end-of-period cash on hand (Corollary 3.1), so C_t^P and C_t^I aggregate consumption over the two occupations. Worker consumption is wage income plus the transfer, $C_t^W = (1-\alpha)Y_t + T_t$, and the transfer is the government budget with zero purchases: the government collects taxes, repays the maturing bond, sells the new issue at the period's settlement, and rebates the balance,

$$T_t = \tau\alpha Y_t - rB_t + B_{t+1}. \tag{35}$$

No government spending term appears in the identity because every fiscal flow is a transfer: the repayment is financed by taxes within the period, and the proceeds of the new issue, raised from capitalist saving at the settlement stage, are consumed by workers out of the same period's output. Substituting C_t^W isolates the rebate as its own absorption line, $Y_t = (1-\alpha)Y_t + C_t^P + C_t^I + I_t + T_t$, splitting worker consumption into its wage financed and transfer financed parts. Appendix B.7 derives each flow.

Next, we define a recursive equilibrium with the state variable being the aggregate capitalist wealth.

Definition 3.1 (Recursive equilibrium)**.** Given the fiscal parameters (τ, φ) and the financing capacity ξ, a *recursive equilibrium* with aggregate capitalist wealth W as the aggregate state consists of value functions V and U, a common saving policy ς, price functions $r(W)$, $p(W)$, and $w(W,\varepsilon)$, allocations $\bar{z}(W)$ and $\theta(W)$, and a law of motion $W' = \Gamma(W,\varepsilon)$ such that at every W:

1. *(Producer optimality)* each producer maximizes his after-tax profits by choosing labor and equity issuance.

2. *(Investor optimality)* each investor chooses the portfolio share θ in the equity index and the risk-free bond that maximizes $CE^{\Lambda}(\mathcal{R}^I)$, with the weight $\Lambda_t = \Lambda(\Gamma(W,\varepsilon))$ of Proposition 3.2;

3. *(Pricing consistency)* the equity price satisfies $q(z) = \mathbb{E}[\mathcal{M}\,\mathcal{R}^K(z,\varepsilon)]$, with $\mathcal{M}$ being the investor's stochastic discount factor of Proposition 3.3;

4. *(Occupational indifference)* the marginal capitalist is indifferent between the two occupations, $CE^{\Lambda}\big(\mathcal{R}^P(\bar{z},\varepsilon)\big) = CE^{\Lambda}\big(\mathcal{R}^I(\varepsilon)\big)$;

5. *(Fiscal rule)* bond supply follows $rB = \varphi\,\mathbb{E}[\tau\Pi^A]$;

6. *(Market clearing)* the equity market, the bond market, and the labor market clear state by state:

$$\theta\,W^I = \xi\,W^P, \qquad (1-\theta)\,W^I = B, \qquad \left(\frac{1-\alpha}{w(\varepsilon)}\right)^{1/\alpha} \exp(\varepsilon)\,\tilde{K} = 1 \quad \text{for every } \varepsilon. \tag{36}$$

7. *(Law of motion for state variable)* next period's aggregate wealth is the saved fraction of aggregate cash on hand, $M(W,\varepsilon) = (1-\tau)\alpha Y + (1-\delta)K + rB$ (Appendix B.7), at the saving

rate evaluated at that wealth:

$$\Gamma(W,\varepsilon) = \varsigma\big(\Gamma(W,\varepsilon)\big)\, M(W,\varepsilon) \quad \text{for every } \varepsilon, \tag{37}$$

where the saving policy and the weights are those of Proposition 3.2 and Corollary 3.1 at the equilibrium returns.

The goods market clears by Walras' law. Given W and the weight function Λ, conditions 1 to 6 form the within-period block: a price system $(r, p, w(\varepsilon))$ and allocations $(\bar{z}, \theta)$ that clear the markets when returns are valued under the shock distribution tilted by Λ_t. Condition 7 regulates the recursion: the saving rate given the state variable and aggregate shock determines the weights that value this period's claims, so the weight function is a fixed point. When the weights are evaluated at a fixed level of aggregate wealth, and the within-period block is the static equilibrium for a constant weight.

4 The Full Two Country Model

To understand how cross border capital flow shope TFP, we extend the model of Section 3 to two countries $c \in \{H, F\}$ whose investors trade cross border assets. Each country is populated by workers, capitalists, and a government, exactly as in the closed economy. The technology and preference parameters $(\alpha, \delta, \gamma, \beta)$ are common across countries; the policy and country primitives $(\tau_c, \varphi_c, \xi_c, \sigma_c, \eta_c)$, together with the openness wedges introduced below, may differ. Country-c producers draw their own aggregate shock ε_c, with $(\varepsilon_H, \varepsilon_F)$ jointly normal, variances (σ_H^2, σ_F^2), and correlation $\nu < 1$. With a single homogeneous consumption good, there are no terms of trade or real exchange rate movements, and the two countries interact only through asset markets. Each country otherwise follows the closed economy structure described above. We therefore focus here on the new elements introduced by financial integration: cross border asset trade, home bias, and the resulting external asset positions.

The aggregate state is now the pair of aggregate capitalist wealths (W_H, W_F), with a law of motion $W_{c,t+1} = \Gamma_c(W_{H,t}, W_{F,t}, \varepsilon_{H,t}, \varepsilon_{F,t})$ for each country, which capitalists take as given. The recursive structure of Section 3 carries over country by country. Proposition 3.2 applies to country-c capitalists with the world state variables (W_H, W_F), and each country has its own common saving rate ς_c and its own continuation weight $\Lambda_{c,t} = \Lambda_c(W_{H,t+1}, W_{F,t+1}) = \big(1 - \varsigma_c(W_{H,t+1}, W_{F,t+1})\big)^{-\gamma}$. The weights differ across countries because the investors from different countries can hold different portfolios and face different returns, and each continutation weight depends on both countries' shocks.

Assets and the home-bias friction Investors trade three asset: a risk-free bond, Home and Foreign risky equity indices, Each country issues its own government bond and its own equity index. The bonds are perfect substitutes while the two equity indices are distinct assets that load on different country aggregate shocks.

Home bias applies to cross border equity holdings and is modeled as an iceberg cost. A non-resident investing in country-a equity receives net return $\mathcal{R}_a^e - \chi_a$, where $\chi_a \geq 0$ is a real resource loss. This wedge captures barriers to international equity investment, such as asymmetric information or capital controls. Because it enters as a haircut to returns, the model preserves the homotheticity. Let θ_a^c denote the share of a country-c investor allocated to country-a equity. The two investors' gross returns are

$$\mathcal{R}_H^I = r + \theta_H^H\big(\mathcal{R}_H^e - r\big) + \theta_F^H\big(\mathcal{R}_F^e - \chi_F - r\big), \qquad \mathcal{R}_F^I = r + \theta_H^F\big(\mathcal{R}_H^e - \chi_H - r\big) + \theta_F^F\big(\mathcal{R}_F^e - r\big). \tag{38}$$

Under CRRA and aggregate risk, investors never short either equity index, though they may short bonds. When $\chi_H = \chi_F = 0$, the model collapses to a fully integrated world economy.

Domestic pricing and the SDFs Pricing the equity indices, as in the closed economy, requires identifying the marginal holders and their SDFs.

Assumption 4.1 (Domestic investors are marginal holders of domestic equity indices)**.**

Under Assumption 4.1, the domestic SDF prices domestic equity:

$$q_c(z) = \mathbb{E}\big[\mathcal{M}_c\, \mathcal{R}_c^K(z, \varepsilon_c)\big], \qquad \mathcal{M}_c = \frac{\Lambda_{c,t}\,(\mathcal{R}_c^I)^{-\gamma}}{\mathbb{E}[\Lambda_{c,t}\,(\mathcal{R}_c^I)^{1-\gamma}]}, \qquad c \in \{H, F\}, \tag{39}$$

with $\Lambda_{c,t}$ the continuation weight of country-c capitalists, the two-country counterpart of Proposition 3.3, and in equilibrium each country's investors hold a strictly positive share of their own equity, $\theta_H^H > 0$ and $\theta_F^F > 0$. All expectations integrate over the joint shock pair $(\varepsilon_H, \varepsilon_F)$: the kernel $\mathcal{M}_c$ inherits the domestic investor's portfolio, so it loads on both countries' shocks whenever cross border equity positions are positive, and the price of a local payoff then reflects its covariance with foreign risk. Foreign investors may also hold domestic equity; when they do, their Euler equation holds at the same asset price but incorporates the cross border wedge. The domestic SDF is useful because it provides the wedge free pricing relation. Foreign demand changes the amount of domestic risk borne by home investors and therefore their SDF. Hence foreign portfolio demand affects q_H through equilibrium risk sharing even though q_H is derived using the domestic pricing kernel. The assumption determines which Euler equation is used to price domestic equity; it does not determine whose demand moves that price in equilibrium.

Assumption 4.1 requires that domestic investors remain active holders of domestic equity as they do not have the cross border wedge faced by foreign investors. This allows the domestic SDF to price the domestic index without insulating its price from foreign demand. As foreign investors absorb more domestic equity, home investors bear less domestic risk, which affects asset prices since the equilibrium covariance between their marginal utility and domestic payoffs changes. The assumption rules out the extreme crossed border holding equilibrium in which foreigners hold all domestic equity despite facing worse terms than residents.

In the bond market, both investors trade bonds without wedges. No-arbitrage condition equalizes the means of the two kernels:

$$\mathbb{E}[\mathcal{M}_H] = \mathbb{E}[\mathcal{M}_F] = \frac{1}{r}, \qquad p_c = \mathbb{E}[\mathcal{M}_c \mathcal{D}_c], \qquad q_c(z) = (1-\tau_c)\, z\, p_c + \frac{1-\delta}{r}. \tag{40}$$

The safe component of the cost of capital is a world price, the risky component a country price, so cross country gaps in the cost of capital are gaps in the pricing of risk. Subtracting the home investor's unwedged first order condition for home equity from the foreign investor's wedged one,

$$\mathbb{E}[\mathcal{M}_F\, \mathcal{R}^e_H] - \mathbb{E}[\mathcal{M}_H\, \mathcal{R}^e_H] = \frac{\chi_H}{r}, \tag{41}$$

and symmetrically for Foreign equity. When $\chi_H = 0$, investors with interior positions must assign the same price to every commonly traded payoff; with a positive wedge, differences in SDF-implied valuations can persist because exploiting them is costly. The wedge is primitive, while the valuation gap is its equilibrium counterpart. As the wedge rises, foreign demand eventually reaches zero at a finite threshold.

Proposition 4.1 (Finite maximum cost of cross border investment)**.** *For $c \in \{H, F\}$, with $-c$ the other country, let $\mathcal{M}^0_{-c}$ denote the country-$(-c)$ investor's SDF at the corner portfolio with no cross border equity. Cross-holdings of country-c equity vanish, $\theta^{-c}_c = 0$, if and only if the wedge reaches a finite threshold:*

$$\chi_c \;\geq\; \bar{\chi}_c \equiv r\left(\mathbb{E}[\mathcal{M}^0_{-c}\, \mathcal{R}^e_c] - 1\right) = \underbrace{\mathbb{E}[\mathcal{R}^e_c] - r}_{\text{premium at the corner}} + \underbrace{r\ \mathrm{Cov}\left(\mathcal{M}^0_{-c}, \mathcal{R}^e_c\right)}_{\text{comovement penalty}}. \tag{42}$$

The proof is in Appendix B.8. Intuitively, cross border equity holdings vanish once the investment wedge exceeds the marginal value of the first unit of foreign equity. This value reflects both the expected return and the diversification benefit when holding foreign equities. The latter falls with the correlation ν between Home and Foreign shocks: as the shocks become more correlated, foreign equity pays off increasingly similar in the same states as the investor's domestic portfolio and therefore provides less insurance. The cutoff wedge is consequently finite and declines in ν.

Equilibrium and net foreign assets Let W_c denote aggregate capitalist wealth in country c, split by the occupational cutoff into investor and producer wealth, $W^I_c = (1 - \bar{z}_c^{-\eta}) W_c$ and $W^P_c = \bar{z}_c^{-\eta} W_c$. Given the pair (W_H, W_F) and the two weight functions, a within-period equilibrium is a price vector (r, p_H, p_F), a pair of cutoffs $(\bar{z}_H, \bar{z}_F)$ and the allocation of portfolios $\theta^H_H, \theta^H_F, \theta^F_H, \theta^F_F$, such that, with the four portfolio weights set by the investors' weighted first order conditions, the three asset markets clear and the two occupational-indifference conditions, $CE^{\Lambda_c}\big(\mathcal{R}^P_c(\bar{z}_c)\big) = CE^{\Lambda_c}\big(\mathcal{R}^I_c\big)$ for $c \in \{H, F\}$, hold. The two equity markets and the world bond market clear:

$$\theta^H_H W^I_H + \theta^F_H W^I_F = \xi_H W^P_H, \qquad \theta^H_F W^I_H + \theta^F_F W^I_F = \xi_F W^P_F, \tag{43}$$

$$\big(1-\theta_H^H-\theta_F^H\big)W_H^I+\big(1-\theta_H^F-\theta_F^F\big)W_F^I=B_H+B_F, \tag{44}$$

with the goods market clearing by Walras' law. The two-country recursive equilibrium is Definition 3.1 applied country by country with the world state varaibles. Each country's saving policy and continuiation weight follow from Proposition 3.2 and Corollary 3.1 at its own returns, and the aggregate wealth dynamic is regulated by $W_{c,t+1}=\varsigma_c(W_{H,t+1},W_{F,t+1})\,M_{c,t}$, with $M_{c,t}$ the cash on hand defined below.

The net foreign asset position of each country follows. Home residents hold $\theta_F^H W_H^I$ of Foreign equity and $(1-\theta_H^H-\theta_F^H)W_H^I$ of bonds; foreigners hold $\theta_H^F W_F^I$ of Home equity; and Home has issued B_H. Collecting these,

$$\mathrm{NFA}_H=\underbrace{\theta_F^H W_H^I-\theta_H^F W_F^I}_{\text{net equity position}}+\underbrace{\big(1-\theta_H^H-\theta_F^H\big)W_H^I-B_H}_{\text{net debt position}}=W_H^I-\big(\xi_H W_H^P+B_H\big), \tag{45}$$

Home's net foreign position is its saving minus the value of the traded assets it issues at home; a country that saves more than it issues is a net creditor.

The two-country equilibrium inherits the recursive structure of the closed economy. Under equity market segmentation, bonds are the only assets traded across countries, and world bond clearing determines the common risk-free rate as

$$r=R_W(\bar z_H,\bar z_F), \tag{46}$$

with R_W decreasing in each country's cutoff. Conditional on r, the two country blocks separate: each country's portfolio condition and occupational indifference condition coincide with the closed economy system of Section 3, except that the domestic bond market-clearing condition is replaced by world bond clearing. These conditions determine a cutoff $\bar z_c(r)$ for each country. The world equilibrium can therefore be characterized by the fixed-point condition

$$R_W\left(\bar z_H(r),\bar z_F(r)\right)=r. \tag{47}$$

Equivalently, substituting $r=R_W(\bar z_H,\bar z_F)$ into the two occupational indifference conditions yields a system of two equations in $\bar z_H,\bar z_F$.

The two-country model introduces one possibility absent in the closed economy. In autarky, domestic investors must jointly absorb the equity issued by producers and the bonds issued by their government. Their equity position cannot exceed their wealth, which requires

$$\bar z_c\geq z_{\theta,c}\equiv(1+\xi_c)^{1/\eta} \tag{48}$$

At this threshold, investors allocate all of their wealth to equity and bond demand falls to zero. In an integrated bond market, this restriction needs to hold only globally. Investors in one country may finance part of their domestic equity position by shorting the world bond, while investors in the

other country absorb both governments' debt. The cutoff in the borrowing country can therefore fall below $z_{\theta,c}$. At most one country can be in this leveraged position, since aggregate world bond holdings must equal the positive supply of government debt. Such leverage arises only when safe assets are sufficiently scarce.

Net foreign income and wealth accumulation External positions affect wealth accumulation. Relative to the closed economy, residents additionally earn income on foreign assets and pay income to foreign holders of domestic assets. Define country c's net foreign investment income as

$$\mathrm{NFI}_{c,t} = \underbrace{\theta^c_{-c}\big(\mathcal{R}^e_{-c} - \chi_{-c}\big)W^I_{c,t} - \theta^{-c}_c\,\mathcal{R}^e_c\,W^I_{-c,t}}_{\text{net equity income}} + \underbrace{r\big[\big(1-\theta^c_c-\theta^c_{-c}\big)W^I_{c,t} - B_{c,t}\big]}_{\text{interest on the net bond position}}, \tag{49}$$

where $-c$ denotes the other country. The first part is the net payoff on foreign equity held by residents minus the payoff to foreign holders of domestic equity. The second is interest on residents' bond holdings net of domestic bond issuance; it can be negative when the country borrows from the rest of the world.

Aggregate capitalist cash on hand can then be written as

$$M_{c,t} = (1-\tau_c)\alpha Y_{c,t} + (1-\delta)K_{c,t} + rB_{c,t} + \mathrm{NFI}_{c,t}. \tag{50}$$

The first three terms are exactly the closed economy cash flow, while $\mathrm{NFI}_{c,t}$ captures the additional income generated by the country's external balance sheet. Next period's wealth is the saved fraction of cash on hand, at the saving rate evaluated at the world state $W_{H,t}, W_{F,t}$,

$$W_{c,t+1} = \varsigma_c M_{c,t}, \tag{51}$$

Thus, external positions affect wealth accumulation through the income they generate. Returns on foreign claims add to domestic wealth accumulation, while payments to foreign claimholders reduce it. When cross border positions vanish, $\mathrm{NFI}_{c,t} = 0$, the wealth accumulation dynamic collapses to its closed economy counterpart. With bilateral equity holdings, even symmetric countries can experience nonzero realized net foreign income; because cross border equity returns are subject to the iceberg cost, its unconditional mean is lower by the resources lost to the wedge.

5 Quantitative Illustration

This section uses the two-country model of Section 4 to illustrate the mechanisms numerically. We solve the within-period block of equilibrium 3.1 at a symmetric benchmark and change one Home primitive variable at a time, and study how cross-country differences in fiscal capacity, wealth, and other fundamentals shape the composition of net foreign asset positions.

5.1 Parameters and solution method

Solution method. We hold aggregate wealth fixed at its benchmark level in both countries and evaluate the continuation weights at that wealth level, and analyse the static equilibrium. The static equilibrium can be simplified to a system of five equations in five unknowns: the productivity cutoffs $(\bar{z}_H, \bar{z}_F)$, the prices of risk (p_H, p_F), and the world risk-free rate r. We solve the equilibrium conditions using a fixed-point approach and damped Newton method that accommodates nonnegative cross-border holdings.

Calibration. Table 4 reports the baseline parameter values. We discuss each choice in turn.

Table 4: Parameters of the numerical illustration

Parameter	Value	Description
α	0.30	capital share
δ	0.25	depreciation
β	0.90	discount factor; capitalists invest the share $\beta/(1+\beta)$ of a unit endowment
γ	2	relative risk aversion
η	7	Pareto tail of productivity draws
ξ	0.80	financing capacity, outside equity per unit of own wealth
τ	0.25	tax on operating profits
φ	1	pledgeable fraction of expected tax revenue
(μ, σ)	$(-1, 1.2)$	mean and standard deviation of the aggregate shock
ν	0.30	correlation of the two countries' shocks
χ_H, χ_F	0.005	wedge on the return to cross-border equity holdings

Notes: all values are common to the two countries at the benchmark. Each experiment moves one Home parameter and holds every other parameter, including Foreign's, at the benchmark.

The preference and technology parameters are the test values used throughout the theory sections. The capital share $\alpha = 0.3$ and the depreciation rate $\delta = 0.25$ are conventional choices for annual models. The discount factor $\beta = 0.90$ implies an annual rate of time preference of approximately 10%, which is on a larger side of the conventional range of discount factor. Relative risk aversion $\gamma = 2$ is within the range used in the business cycle literature. and $\chi_H = \chi_F = 0.005$ is the wedge on the return to cross-border equity holdings, which gives investors a home bias.

The Pareto shape parameter $\eta = 7$ governs the thickness of the right tail of the productivity distribution. Lower values of η imply more dispersion in productivity and a heavier right tail. Our choice of $\eta = 7$ is consistent with estimates from the firm size distribution literature (Gabaix, 2016) and produces a realistic degree of heterogeneity among producers.

The equity financing constraint $\xi = 0.8$ means that entrepreneurs can raise outside equity up to 80% of their net worth, which implies that they retain a minimum stake of $1/(1+\xi) \approx 56\%$ in their firms. This is in the range of calibrations used in the heterogeneous-producer literature (Buera and Moll, 2015; Buera and Shin, 2017). The effective tax rate $\tau = 0.25$ represents a typical effective corporate tax rate for advanced economies. In the comparative statics below, we vary τ across countries to study the effects of fiscal capacity differences.

The aggregate shock parameters $(\mu_\varepsilon, \sigma_\varepsilon) = (-1, 1.2)$ generate meaningful variation in equity returns and ensure that the risk premium is large enough for portfolio investors to hold a mix of bonds and equity. The negative mean $\mu_\varepsilon = -1$ is a normalization that, combined with the lognormal specification, produces realistic expected profit levels.

5.2 Fiscal Capacity and the Composition of External Positions

The main quantitative exercise varies the home country's fiscal capacity, parameterized by τ, starting from the symmetric benchmark where $\tau = \tau^* = 0.25$. Proposition **??** predicts that higher home fiscal capacity generates opposite-signed net bond and net equity positions. We now verify this prediction numerically and trace the mechanism through its component channels.

Net foreign asset positions. Figure 4 reports the central comparative static. As the home tax parameter τ increases above the symmetric benchmark, the home country becomes a net debtor in bonds ($\text{NFA}^B < 0$) and a net creditor in equity ($\text{NFA}^E > 0$). The opposite-signed pattern emerges smoothly and widens as the fiscal capacity asymmetry grows.

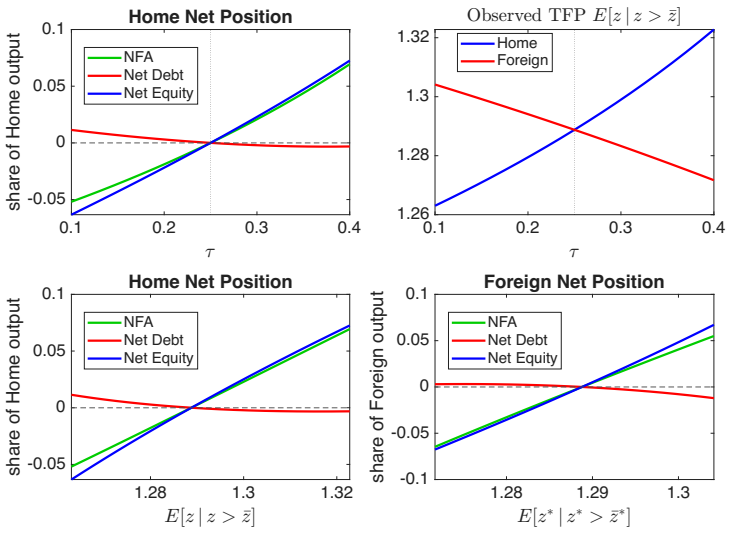
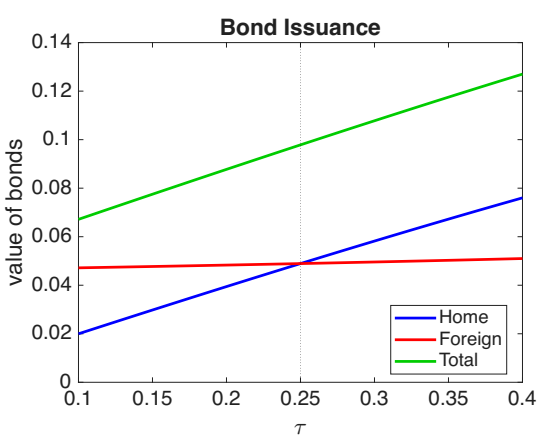


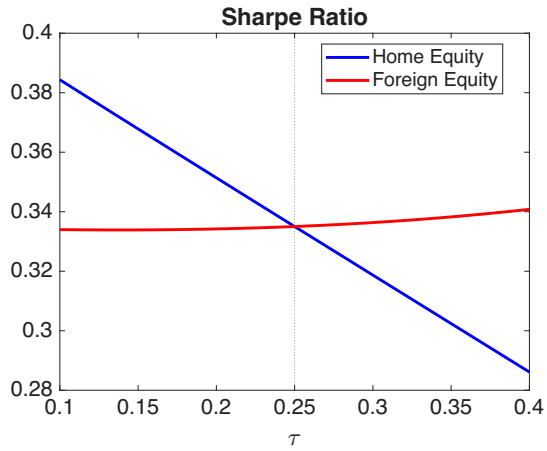
Figure 4: NFA vs. TFP: Varying τ

Notes: The charts in the top row show Home NFA positions and observed TFP when varying τ from the symmetric benchmark. The charts in the bottom row show the Home and Foreign NFA positions with respect to the observed TFP. Positive (negative) values indicate being a net creditor (issuer).

Direct effects: bond supply and risky returns. Figure 5 illustrates the two direct channel. Panel (a) shows that home government bond supply increases monotonically with τ, as higher expected tax revenue allows the government to credibly issue more bonds under the fiscal rule. Panel (b) shows that the Sharpe ratio of home equity declines with τ,reflecting the reduction in after-tax profits. The combination of increased bond supply and reduced equity attractiveness is the fundamental force driving the composition of external positions.



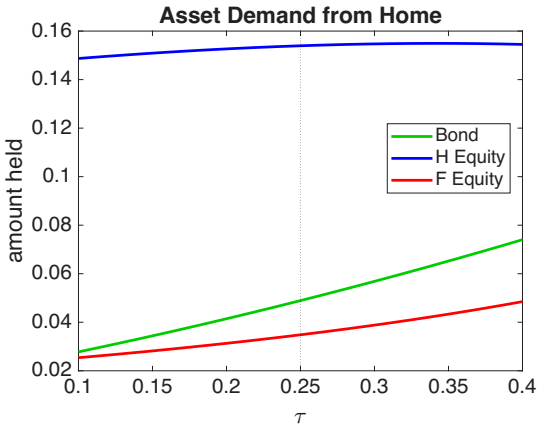
(a) Bond supply



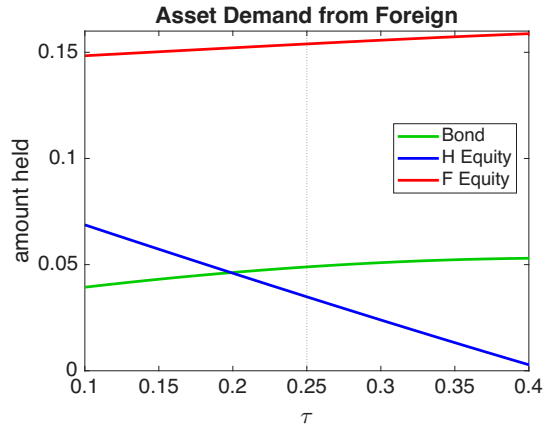
(b) Sharpe ratios

Figure 5: Direct effects of home fiscal capacity: bond supply and equity Sharpe ratios.

Portfolio rebalancing. Figure 6 traces the portfolio rebalancing channel. Panel (a) shows the asset demands of home non-producing entrepreneurs. When τ rises, demand for foreign equity and bonds increases. Panel (b) shows the asset demands of foreign non-producing entrepreneurs. They also reduce their holdings of home equity (due to lower Sharpe ratio) and increase bond holdings. The portfolio rebalancing is symmetric in direction, with both home and foreign investors shifting away from risky assets. However, the rebalancing is asymmetric in magnitude, since home investors increase their foreign equity demand while foreign investors do not have a new attractive risky alternative.



(a) Home portfolio demands



(b) Foreign portfolio demands

Figure 6: Portfolio rebalancing: asset demands as a function of home fiscal capacity τ.

Selection and observed TFP. The selection channel is crucial for mapping the model to the empirical evidence. Higher τ reduces after-tax profitability, forcing marginal home producers to exit (the cutoff $\bar{z}$ rises). Conversely, the lower portfolio return makes foreign production relatively more attractive, so the foreign cutoff $\bar{z}^*$ falls. The net effect is that observed TFP (measured

as $\mathbb{E}[z \mid z \geq \bar{z}]$) rises in the home country and falls in the foreign country. The bottom row of Figure 4 plots net foreign asset positions against observed TFP for both countries as τ varies. The home country moves toward higher TFP, negative net bond positions, and positive net equity positions, while the foreign country moves in the opposite direction. The resulting pattern matches the empirical evidence that countries with higher observed TFP are net debtors in bonds and net creditors in equity.

5.3 Additional Comparative Statics

The framework is sufficiently flexible to study how cross-country asymmetries beyond fiscal capacity shape the composition of capital flows. In this section, we consider three additional exercises that vary initial wealth, equity financing constraints and the productivity distribution. Full comparative statics panels are reported in Appendix C.

5.3.1 Initial Wealth

Figure 7 reports net foreign asset positions as a function of home initial wealth A. Unlike the fiscal capacity exercise, varying wealth moves net bond and net equity positions in the same direction. The wealthier countries are net creditors in both asset classes. This is because higher wealth raises demand for all assets without differentially affecting the return on domestic equity relative to bonds. General-equilibrium wage effects raise the cutoff $\bar{z}$ and measured TFP, but there is no force that asymmetrically shifts bond supply relative to equity demand.

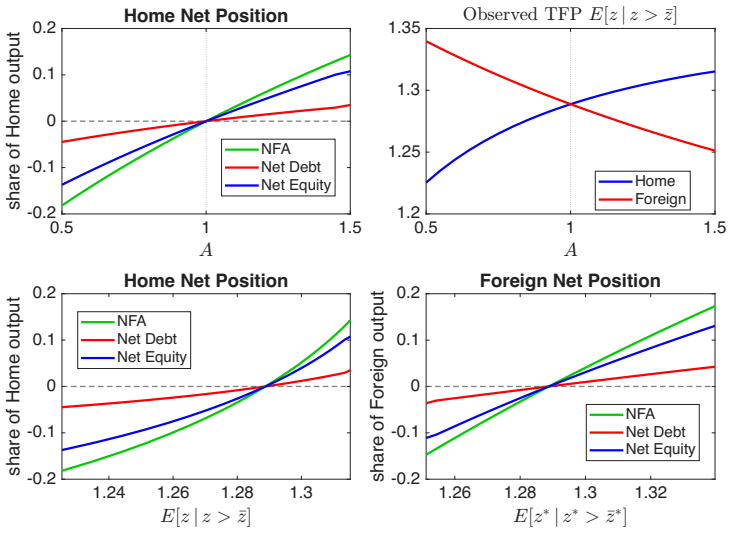


Figure 7: NFA vs. TFP: Varying A

5.3.2 Financial Conditions

Differences in equity financing constraints affect the supply of risky assets and the tightness of domestic equity markets. Figure 8 reports net foreign asset positions as a function of the home

equity constraint parameter ξ. A tighter constraint (lower ξ) reduces the supply of home equity, which in turn affects portfolio demands and equilibrium returns. The effects on NFA composition depend on the interaction between reduced equity supply and the endogenous adjustment of the cutoff and risk-free rate.

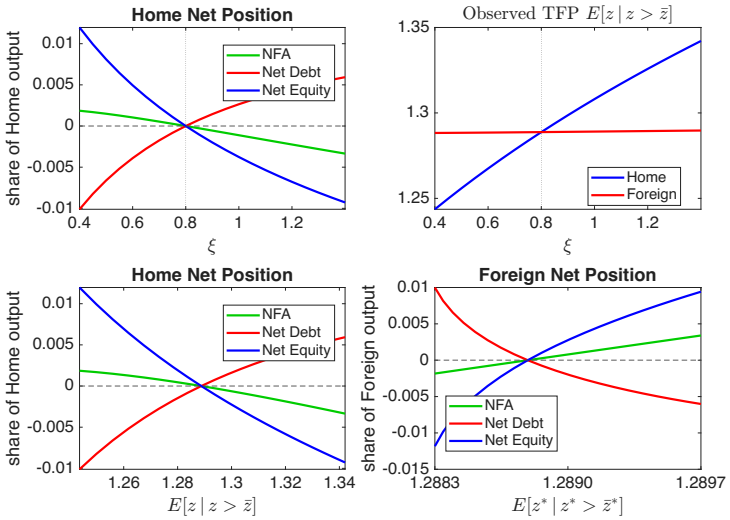


Figure 8: NFA vs. TFP: Varying ξ

This exercise connects to the literature on financial development and capital flows. Caballero et al. (2008) and Mendoza et al. (2009) emphasize that differences in the ability to produce financial assets can drive global imbalances. In our framework, the equity constraint ξ directly influences the supply of equity, and its effects on NFA composition depend on the interaction with the fiscal capacity channel.

5.3.3 Productivity Distribution

Differences in the Pareto shape parameter η capture variation in the dispersion of entrepreneurial talent across countries. A lower η implies a thicker right tail, meaning there are more highly productive entrepreneurs. This affects both the cutoff and the average productivity among active producers. Figure 9 reports the NFA decomposition as η varies for the home country.

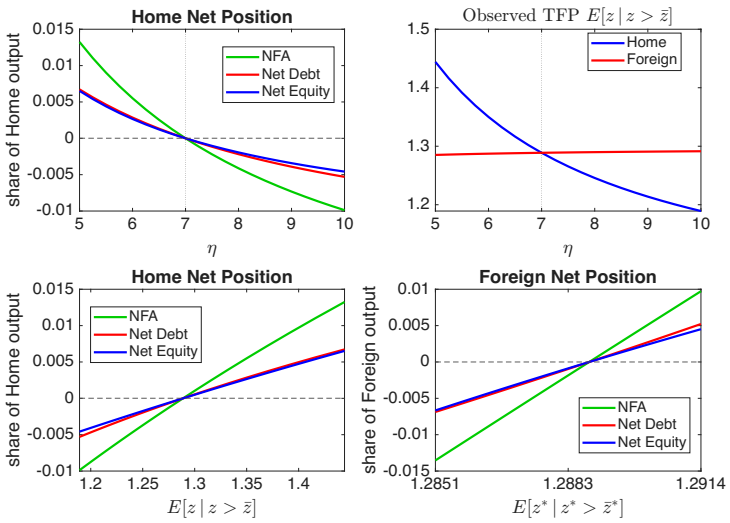


Figure 9: NFA vs. TFP: Varying η

The Pareto shape parameter governs how concentrated productive capacity is among a few highly productive entrepreneurs versus spread across many. A thicker right tail (lower η) means more entrepreneurs draw high productivity, which increases equity supply and raises the share of aggregate capital deployed by the most productive firms. This in turn affects the sensitivity of the cutoff and equity returns to changes in equilibrium prices, both of which feed into the composition of external positions.

6 Conclusion

This paper revisits the allocation puzzle (Gourinchas and Jeanne, 2013) by decomposing global capital flows into risky and safe components. Using data from the External Wealth of Nations database and the Penn World Table for 43 countries over the period 1970–2023, we document that the allocation puzzle is primarily a debt puzzle. Equity inflows move toward faster-growing economies, consistent with neoclassical theory, while debt inflows move in the opposite direction. The same composition appears in net external positions, with high-TFP countries holding positive net equity positions and negative net debt positions.

To interpret these facts, we develop a model with heterogeneous entrepreneurs, equity financing constraints, government bond issuance, and international portfolio choice. The key mechanism in the model operates through fiscal capacity. Countries with stronger fiscal capacity issue more government bonds, and the higher effective tax rate depresses after-tax equity returns. This triggers a selection effect that raises observed TFP by crowding out less productive producers. The resulting portfolio rebalancing generates opposing flows of debt and equity positions across countries. Beyond the focus on fiscal capacity, this model serves as a more general framework for examining how economic fundamentals may affect the joint allocation of risky and safe capital, which consequently shapes global capital flows and economic outcomes.

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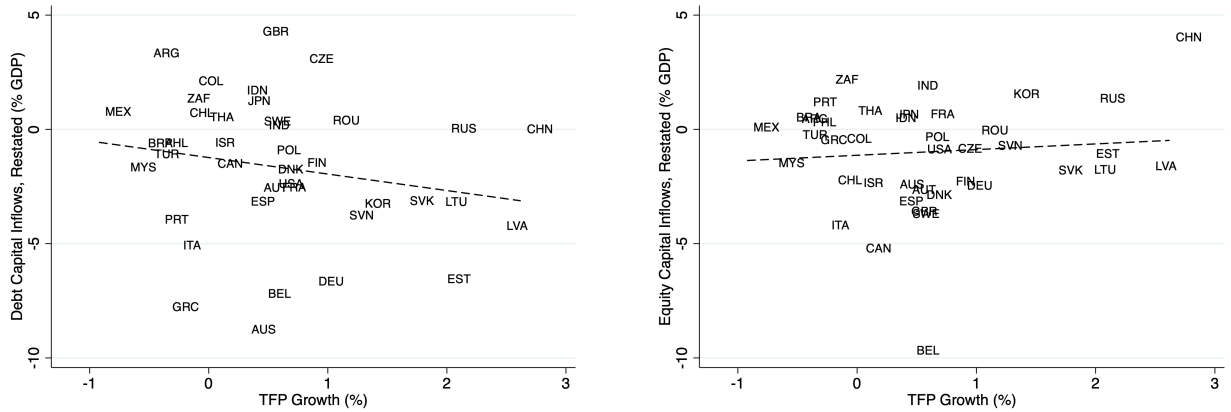
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A Additional Empirical Results

A.1 Cross-country Plots

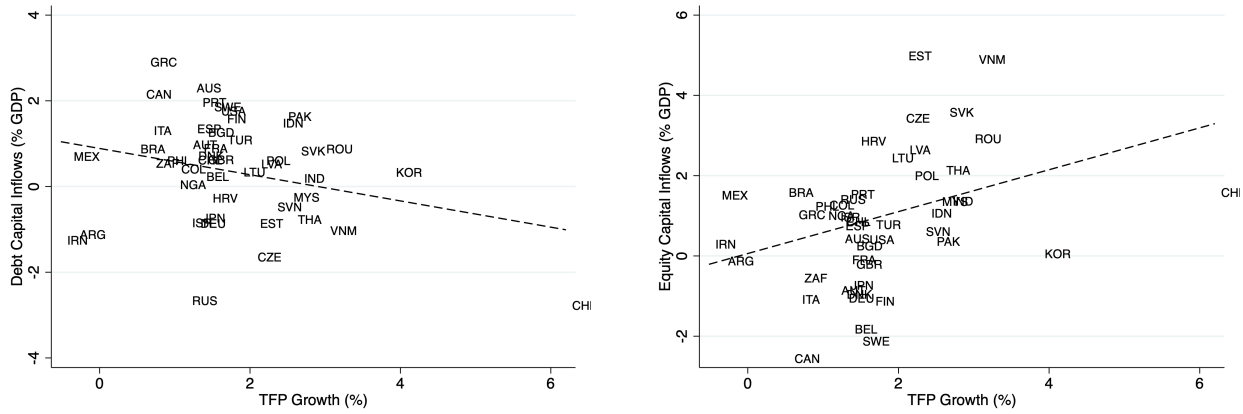
We present alternative specifications of the cross-country scatterplots.



(a) Debt inflows (restated)

(b) Equity inflows (restated)

Figure A.1: Capital inflows and TFP growth using restated data.



(a) Debt inflows

(b) Equity inflows

Figure A.2: Capital inflows and TFP growth using TFP measurements from Gourinchas and Jeanne (2013).

A.2 Cross-country regressions

	Net debt assets					Net equity assets				
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
TFP level (std.)	-5.090**	-4.401	-4.171	-2.466	-3.129	8.121***	5.086**	8.116***	3.193	3.242
	(2.328)	(3.126)	(2.901)	(3.954)	(3.661)	(1.956)	(2.077)	(2.575)	(2.290)	(2.405)
Tax from income/profits (% of GDP)		-0.669			-0.362		1.880***			1.345**
		(0.503)			(0.834)		(0.436)			(0.538)
Fiscal capacity (std.)			-5.221*		-4.515			0.889		-2.573
			(3.038)		(4.378)			(2.045)		(1.824)
State capacity (std.)				-7.349	-1.427				12.240***	8.207
				(4.907)	(9.657)				(3.556)	(5.053)
Human capital index (std.)		6.766*	7.238**	9.116**	8.182*		-4.997	-2.651	-7.859*	-7.227*
		(3.602)	(3.021)	(3.974)	(4.370)		(3.075)	(3.811)	(4.572)	(3.914)
Chinn-Ito index (std.)		-2.867	-2.715	-2.353	-2.262		-0.384	1.236	-0.526	-0.773
		(3.288)	(3.398)	(3.353)	(3.458)		(3.154)	(3.559)	(3.440)	(3.082)
Countries	43	43	43	43	43	43	43	43	43	43

Table A.1: Cross-country regressions: all controls included jointly

	Net debt assets				Net equity assets			
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
TFP level (std.)	-5.090**				8.121***			
	(2.328)				(1.956)			
Tax from income/profits (% of GDP)		-0.609				1.735***		
		(0.379)				(0.320)		
Fiscal capacity (std.)			-4.691**				4.232*	
			(2.299)				(2.364)	
State capacity (std.)				-2.899				7.337***
				(2.238)				(1.894)
Countries	43	46	46	46	43	46	46	46

Table A.2: Cross-country regressions: country fundamentals without TFP

Notes: The dependent variable in each column is the corresponding net foreign asset measured as a percentage of GDP. TFP level is normalized to have a mean of zero and a unit standard deviation. Robust standard errors are reported in parentheses. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.1$.

A.3 Panel regressions

	(1) Total Cap. Inflow	(2) Debt Cap. Inflow	(3) Equity Cap. Inflow
TFP Growth	-0.022	-0.186*	0.389***
	(0.083)	(0.107)	(0.108)
Observations	1143	1143	1143
Country FE	Yes	Yes	Yes
Year FE	Yes	Yes	Yes

Table A.3: Capital inflows vs. TFP growth: Post-1995 sample

	(1) Total Cap. Inflow	(2) Debt Cap. Inflow	(3) Equity Cap. Inflow
TFP Growth	-0.106	-0.236**	0.138
	(0.067)	(0.092)	(0.094)
Observations	660	660	660
Country FE	Yes	Yes	Yes
Year FE	Yes	Yes	Yes

Table A.4: Capital inflows vs. TFP growth: Post-2007 sample

	(1) Total Cap. Inflow	(2) Debt Cap. Inflow	(3) Equity Cap. Inflow
TFP Growth	-0.121*	-0.186***	0.193***
	(0.062)	(0.054)	(0.070)
Observations	1156	1156	1156
Country FE	Yes	Yes	Yes
Year FE	Yes	Yes	Yes

Table A.5: Capital inflows vs. TFP growth: Pre-2008 sample

Notes: The dependent variable in each column is the corresponding capital inflow measured as a percentage of GDP. TFP growth is the annual percentage change in TFP from the Penn World Table. All specifications include country and year fixed effects. Standard errors, clustered at the country level, are reported in parentheses. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.1$.

	(1) Total Cap. Inflow	(2) Debt Cap. Inflow	(3) Equity Cap. Inflow
TFP Growth	-0.093	-0.158***	0.235***
	(0.061)	(0.049)	(0.069)
Observations	1652	1652	1652
Country FE	Yes	Yes	Yes
Year FE	Yes	Yes	Yes

Table A.6: Capital inflows vs. TFP growth: Pre-2020 sample

	(1) Total Cap. Inflow	(2) Debt Cap. Inflow	(3) Equity Cap. Inflow
TFP Growth	-0.121***	-0.204***	0.113
	(0.040)	(0.060)	(0.077)
Observations	2267	2267	2267
Country FE	Yes	Yes	Yes
Year FE	Yes	Yes	Yes

Table A.7: Capital inflows vs. TFP growth: Gourinchas and Jeanne (2013) sample

	(1) Total Cap. Inflow	(2) Debt Cap. Inflow	(3) Equity Cap. Inflow
TFP Growth	-0.114	-0.211***	0.281***
	(0.072)	(0.063)	(0.061)
Observations	1608	1608	1608
Country FE	Yes	Yes	Yes
Year FE	Yes	Yes	Yes

Table A.8: Capital inflows vs. TFP growth: Excluding extrema

Notes: The dependent variable in each column is the corresponding capital inflow measured as a percentage of GDP. TFP growth is the annual percentage change in TFP from the Penn World Table. All specifications include country and year fixed effects. Standard errors, clustered at the country level, are reported in parentheses. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.1$.

	(1) Aggr.	(2) Debt	(3) Equity	(4) Portf. Debt	(5) Reserves	(6) Portf. Equity	(7) FDI
TFP Growth	-0.089	-0.186***	0.234***	-0.053	-0.133***	0.131***	0.109***
	(0.078)	(0.061)	(0.062)	(0.057)	(0.034)	(0.045)	(0.039)
Observations	1816	1816	1816	1816	1816	1816	1816
Country FE	No	No	No	No	No	No	No
Year FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes

Table A.9: Capital inflows vs. TFP growth with no country fixed effects

	(1) Aggr.	(2) Debt	(3) Equity	(4) Portf. Debt	(5) Reserves	(6) Portf. Equity	(7) FDI
TFP Growth	-0.080	-0.175***	0.204***	-0.050	-0.124***	0.124***	0.084***
	(0.056)	(0.047)	(0.054)	(0.044)	(0.025)	(0.041)	(0.030)
Observations	1855	2046	2018	2046	2071	2021	2037
Country FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes

Table A.10: Capital inflows vs. TFP growth without imposing an overlapping sample

Notes: The dependent variable in each column is the corresponding capital inflow measured as a percentage of GDP. TFP growth is the annual percentage change in TFP from the Penn World Table. All specifications include country and year fixed effects. Standard errors, clustered at the country level, are reported in parentheses. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.1$.

B Proofs and Derivations

B.1 Producer optimization and the cost of capital

Labor-market clearing termines the wage. Define $\tilde{K} \equiv \int_{i:z\geq\bar{z}} zk(z)di$. Then labor-market clearing implies

$$\left(\frac{1-\alpha}{w}\right)^{1/\alpha} \exp(\varepsilon)\tilde{K} = 1 \quad \Longrightarrow \quad w = (1-\alpha)\left(\exp(\varepsilon)\tilde{K}\right)^{\alpha}. \tag{B.1}$$

B.2 Proof of Proposition 3.1

Proof.

By Proposition 3.4, every active producer issues to the financing cap, so for each $i \in \mathcal{P}$ the market value of his outside equity equals the amount raised,

$$(1-s(z))\,q(z)\,k = E = \xi a. \tag{B.2}$$

This has two implications. First, the index price collapses to the levered wealth of producers:

$$P^e = \int_{i\in\mathcal{P}} (1 - s(z))\, k(z)\, q(z)\, di = \xi \int_{i\in\mathcal{P}} a_i\, di = \xi W^P. \tag{B.3}$$

Second, each firm's contribution to the index payoff is its equity value times its equity return,

$$(1 - s(z))\, k(z)\, \mathcal{R}^K(z,\varepsilon) = \big[(1 - s(z))\, q(z)\, k(z)\big]\, \frac{\mathcal{R}^K(z,\varepsilon)}{q(z)} = \xi a_i\, \frac{\mathcal{R}^K(z_i,\varepsilon)}{q(z_i)}, \tag{B.4}$$

$$\text{so} \qquad \Pi^e(\varepsilon) = \xi \int_{i\in\mathcal{P}} a_i\, \frac{\mathcal{R}^K(z_i,\varepsilon)}{q(z_i)}\, di. \tag{B.5}$$

Because the productivity draw is independent of wealth and the occupational choice only depends on z, for any function g, $\int_{i\in\mathcal{P}} a_i\, g(z_i)\, di = \bar{A}\, \mathbb{E}_z\big[g(z)\, \mathbf{1}\{i \in \mathcal{P}\}\big]$, where $\bar{A}$ denotes mean capitalist wealth, we have

$$\mathcal{R}^e(\varepsilon) = \frac{\Pi^e(\varepsilon)}{P^e} = \frac{\mathbb{E}_z\big[\frac{\mathcal{R}^K(z,\varepsilon)}{q(z)}\, \mathbf{1}\{i \in \mathcal{P}\}\big]}{\bar{A}\, \Pr(i \in \mathcal{P})} = \mathbb{E}_z\left[\frac{\mathcal{R}^K(z,\varepsilon)}{q(z)} \;\middle|\; i \in \mathcal{P}\right]. \tag{B.6}$$

For the pricing condition, exchange the order of integration over z and ε (Fubini) and use the definition of the equity price, $q(z) = \mathbb{E}_\varepsilon[\mathcal{M}\, \mathcal{R}^K(z,\varepsilon)]$:

$$\mathbb{E}_\varepsilon\big[\mathcal{M}\, \mathcal{R}^e\big] = \mathbb{E}_z\left[\frac{\mathbb{E}_\varepsilon[\mathcal{M}\, \mathcal{R}^K(z,\varepsilon)]}{q(z)} \;\middle|\; i \in \mathcal{P}\right] = \mathbb{E}_z\left[\frac{q(z)}{q(z)} \;\middle|\; i \in \mathcal{P}\right] = 1 : \tag{B.7}$$

the index is a value-weighted bundle of claims, each of which is individually priced by $\mathcal{M}$, so the bundle is priced as well. □

B.3 Proof of Proposition 3.2

Proof.

The proof has three steps: (i) show that the occupation chosen today has no option value; (ii) derive that the value of each occupation follows from a conjectured form of the value function, which we verify; and (iii) compare the value of two occupations and derive cutoff rule.

Step 1: no option value. The capitalist enters next period with the state $(a', z'; W')$. The draw z' is i.i.d., so its distribution is the same whichever occupation he chooses today, and $W' = \Gamma(W, \varepsilon)$ depends only on current aggregate wealth and the shock. For every a' and W' the expected continuation value $\mathbb{E}_{z'}[V(a', z'; W')]$, and hence the end-of-period value $U(m; W')$, is therefore the same for a capitalist regardless the occupation. The two occupations differ only through the cash on hand $m^o = \mathcal{R}^o a$ that depends on the returns.

Step 2: the value of each occupation. By homothecity of CRRA preference, we have

$$\mathbb{E}\big[\Lambda\, u(\mathcal{R}a)\big] = \mathbb{E}[\Lambda]\, \frac{a^{1-\gamma}}{1-\gamma}\, \mathbb{E}^\Lambda[\mathcal{R}^{1-\gamma}] = \mathbb{E}[\Lambda]\, u\big(CE^\Lambda(\mathcal{R})\, a\big). \tag{B.8}$$

Conjecture the power form for value fuction $V(a,z;W)=\psi(z;W)\,a^{1-\gamma}/(1-\gamma)$ and write $\bar{\psi}(W')\equiv\mathbb{E}_{z'}[\psi(z';W')]$, which is the same for all capitalists. Given realized cash on hand m and the realized state W', the consumption-saving problem for each capitalist is

$$U(m;W')=\max_{0\le a'\le m}\; u(m-a')+\beta\,\bar{\psi}(W')\,\frac{(a')^{1-\gamma}}{1-\gamma}. \tag{B.9}$$

Both terms are strictly concave in a' and the first-order condition $(m-a')^{-\gamma}=\beta\bar{\psi}(W')\,(a')^{-\gamma}$ characterizes the linear policy function $a'=\varsigma(W')\,m$ with $\varsigma(W')=b/(1+b)$ and $b\equiv\big(\beta\bar{\psi}(W')\big)^{1/\gamma}$. Substituting back and using $\beta\bar{\psi}(W')=b^{\gamma}$,

$$\begin{aligned}U(m;W')&=\frac{m^{1-\gamma}}{1-\gamma}\Big[(1+b)^{\gamma-1}+b^{\gamma}\Big(\frac{b}{1+b}\Big)^{1-\gamma}\Big]=\Lambda(W')\,\frac{m^{1-\gamma}}{1-\gamma},\\ \Lambda(W')&=(1+b)^{\gamma}=\big(1-\varsigma(W')\big)^{-\gamma}.\end{aligned} \tag{B.10}$$

The coefficient depends on the state through $\bar{\psi}(W')$ and is common to the two occupations. Since $W_{t+1}=\Gamma(W,\varepsilon)$ is a function of the shock, $\Lambda_t\equiv\Lambda(W_{t+1})$ is a random variable when occupations are chosen. Substituting $m^o=\mathcal{R}^o(z;x)\,a$ and applying the homogeneity identity,

$$\mathbb{E}_{\varepsilon}\big[U(\mathcal{R}^o(z;x)\,a;W_{t+1})\big]=\mathbb{E}[\Lambda_t]\,\frac{CE^{\Lambda}\big(\mathcal{R}^o(z;x)\big)^{1-\gamma}}{1-\gamma}\,a^{1-\gamma}. \tag{B.11}$$

The map $y\mapsto y^{1-\gamma}/(1-\gamma)$ is strictly increasing for every $\gamma>0$, so the financial choice within an occupation maximizes $CE^{\Lambda}(\mathcal{R}^o(z;x))$ over x, and the value of occupation o is the expression (23) of the proposition. For a producer the choice is the issuance decision, whose solution is the corner of Section 3.1 because the continuation weight does not change the state-by-state ranking of payoffs; for an investor it is the portfolio share, which is the solution of the problem in Section 3.3.

Verification: the value function is

$$\begin{aligned}V(a,z;W)&=\max_{o}V^o(a,z;W)=\psi(z;W)\,\frac{a^{1-\gamma}}{1-\gamma},\\ \psi(z;W)&=\mathbb{E}[\Lambda_t]\,\Big(\max_{o}CE^{\Lambda}\big(\mathcal{R}^o(z)\big)\Big)^{1-\gamma}>0,\end{aligned} \tag{B.12}$$

which is a power form in wealth, where the continuation weights and the returns are those of the period with state W. Averaging ψ over the draw gives the continuation coefficient $\bar{\psi}(W)$, and thus the saving policy, the continuation weights, and the value function are determined jointly as a fixed point in the function $\bar{\psi}$; Definition 3.1 imposes this consistency in equilibrium.

Step 3: the comparison and the cutoff. By (23) the two values share the factor $\mathbb{E}[\Lambda_t]\,a^{1-\gamma}/(1-\gamma)$, and each is strictly increasing in the weighted certainty equivalent of its return, so the capitalist produces if and only if $CE^{\Lambda}(\mathcal{R}^P(z,\varepsilon))\ge CE^{\Lambda}(\mathcal{R}^I(\varepsilon))$. The investor's return does not depend on his own z, so the hurdle $H\equiv CE^{\Lambda}(\mathcal{R}^I)$ is common to all capitalists. The producer return $\mathcal{R}^P(z,\varepsilon)=[1+\xi(1-1/q(z))]\,\mathcal{R}^K(z,\varepsilon)$ rises with z in every state, since $\mathcal{R}^K(z,\varepsilon)=(1-\tau)z\mathcal{D}(\varepsilon)+(1-\delta)$ is

strictly increasing in z and the financing multiplier $1+\xi(1-1/q(z))$ is nondecreasing in z. It is equal to one below the issuance threshold and increasing above. The weighted certainty equivalent is an expectation under a probability measure with positive density $\Lambda_t/\mathbb{E}[\Lambda_t]$, so it respects statewise dominance, and $P(z) \equiv CE^{\Lambda}(\mathcal{R}^P(z,\varepsilon))$ is strictly increasing in z while H is flat. The producing set $\{z : P(z) \geq H\}$ is therefore an upper interval $[\bar{z}, \infty)$ whose endpoint satisfies $P(\bar{z}) = H$.

Log utility. With $\gamma = 1$, conjecture $V(a, z; W) = \ln a/(1-\beta) + g(z; W)$ and write $\bar{g}(W') \equiv \mathbb{E}_{z'}[g(z'; W')]$. The first-order condition of the saving problem, $1/(m-a') = \beta/[(1-\beta)a']$, gives $a' = \beta m$ and $U(m; W') = \ln m/(1-\beta) + \ell(W')$ with $\ell(W') = \ln(1-\beta) + \beta \ln \beta/(1-\beta) + \beta\, \bar{g}(W')$. For any choice, $\mathbb{E}_{\varepsilon}[U(\mathcal{R}^o(z;x)a; W_{t+1})] = \{\mathbb{E}[\ln \mathcal{R}^o(z;x)] + \ln a\}/(1-\beta) + \mathbb{E}[\ell(W_{t+1})]$, and the last term is the same for every choice. The financial choice maximizes $\mathbb{E}[\ln \mathcal{R}^o(z;x)]$, the conjecture is verified with $g(z; W) = \max_o \mathbb{E}[\ln \mathcal{R}^o(z)]/(1-\beta) + \mathbb{E}[\ell(W_{t+1})]$, and the capitalist produces if and only if $\mathbb{E}[\ln \mathcal{R}^P(z,\varepsilon)] \geq \mathbb{E}[\ln \mathcal{R}^I(\varepsilon)]$.

Corollary 3.1 is a byproduct of Step 2. The linear policy $a' = \varsigma(W')m$ is the saving rule out of end-of-period cash on hand; its rate is independent of the level m by homotheticity and of the occupation because $\bar{\psi}(W')$ is an expectation over the next draw and the current occupation has been realized before the saving choice.

□

B.4 Proof of Proposition 3.3

Proof.

We prove the proposition in two steps.

Step 1: curveture of SDF. By Proposition 3.2, the investor's end-of-period consumption is $c = (1-\varsigma(W'))\, m$ with $m = \mathcal{R}^I a$, so his marginal utility of consumption is $u'(c) = a^{-\gamma}\, \Lambda(W')\, (\mathcal{R}^I)^{-\gamma}$. The weight $\Lambda(W') = (1-\varsigma(W'))^{-\gamma}$ is random through $W' = \Gamma(W, \varepsilon)$. The first-order condition for any traded asset with return R_x is

$$\mathbb{E}\big[\Lambda_t\, (\mathcal{R}^I)^{-\gamma}\, (R_x - r)\big] = 0, \tag{B.13}$$

and excess returns are priced by any kernel proportional to $\Lambda_t(\mathcal{R}^I)^{-\gamma}$.

Step 2: level of SDF. The level is determined by the investor's own portfolio. One unit of wealth allocated at his own weights costs one unit and pays $\mathcal{R}^I$, so a valid kernel must satisfy $\mathbb{E}[\mathcal{M}\mathcal{R}^I] = 1$. Writing $\mathcal{M} = \lambda\, \Lambda_t(\mathcal{R}^I)^{-\gamma}$ and solving,

$$\lambda\, \mathbb{E}[\Lambda_t(\mathcal{R}^I)^{1-\gamma}] = 1 \quad \Longrightarrow \quad \mathcal{M} = \frac{\Lambda_t\, (\mathcal{R}^I)^{-\gamma}}{\mathbb{E}[\Lambda_t\, (\mathcal{R}^I)^{1-\gamma}]}. \tag{B.14}$$

Together with the portfolio first-order condition, the normalization generates the whole pricing system

$$1 = \mathbb{E}[\mathcal{M}\mathcal{R}^I] = r\, \mathbb{E}[\mathcal{M}] + \theta\, \mathbb{E}\big[\mathcal{M}\, (\mathcal{R}^e - r)\big] \tag{B.15}$$

and using the first-order condition to eliminate the excess term leaves $\mathbb{E}[\mathcal{M}] = 1/r$; the equity first-order condition then implies $\mathbb{E}[\mathcal{M}\mathcal{R}^e] = 1$. Finally, an interior holder's Euler equation prices every claim he trades, so the price of a claim to the per-unit capital payoff is $q(z) = \mathbb{E}[\mathcal{M}\,\mathcal{R}^K(z,\varepsilon)]$. □

B.5 Proof of Proposition 3.4

We prove the proposition in a form that covers both the closed economy of Section 3 and the two country model of Section 4. Throughout, $\mathcal{M}_c$ denotes the kernel of the investors who both price the country's equity and define the production hurdle. Write $P(z) \equiv CE\big(\mathcal{R}^P(z,\varepsilon)\big)$ for the certainty-equivalent return on production and $H \equiv CE(\mathcal{R}^I)$ for the hurdle. When $r > 1-\delta$, the issuance threshold solves $q(z_q) = 1$ in closed form,

$$z_q = \frac{1 - \frac{1-\delta}{r}}{(1-\tau)\,p}; \tag{B.16}$$

if instead $r \le 1-\delta$ or $q(1) \ge 1$, then $q(z) > 1$ for every type, all producers issue to the cap, and the proposition is immediate. Throughout this proof, expectations over the shock are taken under the distribution tilted by the continuation weight Λ_t of Proposition 3.2, so CE stands for CE^{Λ} and $\mathcal{M}$ for the weighted kernel of Proposition 3.3, which under the tilted distribution is $(\mathcal{R}^I)^{-\gamma}/\mathbb{E}^{\Lambda}[(\mathcal{R}^I)^{1-\gamma}]$. The argument runs through a dominance lemma.

Lemma B.1 (Dominance of the investor's portfolio)**.** *Let $\mathcal{R}^I$ be the investor's portfolio return and $\mathcal{M} = (\mathcal{R}^I)^{-\gamma}/\mathbb{E}[(\mathcal{R}^I)^{1-\gamma}]$ its normalized kernel, so that $\mathbb{E}[\mathcal{M}\mathcal{R}^I] = 1$ holds identically. Then for any non-negative payoff X with $\mathbb{E}[\mathcal{M}X] \le 1$,*

$$CE(X) \le CE(\mathcal{R}^I), \tag{B.17}$$

with equality if and only if $X = \mathcal{R}^I$ almost surely.

Proof.
With CRRA utility, $U'(W) = W^{-\gamma}$, so marginal utility at the portfolio return is proportional to the kernel by construction:

$$U'(\mathcal{R}^I) = (\mathcal{R}^I)^{-\gamma} = K\,\mathcal{M}, \qquad K \equiv \mathbb{E}[(\mathcal{R}^I)^{1-\gamma}] > 0. \tag{B.18}$$

Concavity puts U below every tangent, $U(x) \le U(y) + U'(y)(x-y)$. Applying this state by state with $x = X$ and $y = \mathcal{R}^I$ and taking expectations,

$$\mathbb{E}[U(X)] \le \mathbb{E}[U(\mathcal{R}^I)] + K\big(\underbrace{\mathbb{E}[\mathcal{M}X]}_{\le 1} - \underbrace{\mathbb{E}[\mathcal{M}\mathcal{R}^I]}_{=1}\big) \le \mathbb{E}[U(\mathcal{R}^I)]. \tag{B.19}$$

The first inequality is implied by Jensen's inequality. The payoff X costs weakly less than the portfolio, and marginal utility prices the difference. Since $CE = U^{-1}(\mathbb{E}[U(\cdot)])$ and U^{-1} is increasing,

the ranking transfers to certainty equivalents. Equality requires the supporting-line inequality to bind state by state, and strict concavity then implies $X = \mathcal{R}^I$ almost surely. Under log utility the whole argument is an application of Jensen's inequality, where $\mathbb{E}[\ln X] - \mathbb{E}[\ln \mathcal{R}^I] = \mathbb{E}[\ln(\mathcal{M}X)] \leq \ln \mathbb{E}[\mathcal{M}X] \leq 0$. □

Proof of Proposition 3.4.
At $z = z_q$, $q(z_q) = 1$, so the issuance multiplier equals one for any $E \in [0, \xi a]$ and

$$\mathcal{R}^P(z_q, \varepsilon) = \Big[1 + \frac{E}{a}\Big(1 - \frac{1}{q(z_q)}\Big)\Big]\mathcal{R}^K(z_q, \varepsilon) = \mathcal{R}^K(z_q, \varepsilon) : \tag{B.20}$$

selling equity at par neither creates nor destroys value for the seller, so producing at the threshold is owning the raw capital payoff.

The investors are the unwedged marginal holders of the country's equity, so

$$\mathbb{E}\big[\mathcal{M}\,\mathcal{R}^K(z_q, \varepsilon)\big] = q(z_q) = 1. \tag{B.21}$$

The marginal-at-issuance firm's payoff is exactly a unit-cost claim under the kernel of the agents whose outside option defines the hurdle. Taking $X = \mathcal{R}^K(z_q, \varepsilon)$,

$$P(z_q) = CE\big(\mathcal{R}^K(z_q, \varepsilon)\big) \leq CE\big(\mathcal{R}^I\big) = H. \tag{B.22}$$

The inequality is strict unless the firm at the issuance threshold gives exactly the same return as the investor's optimal portfolio in almost every aggregate state.

Moreover, $\mathcal{R}^K(z, \varepsilon) = (1-\tau)z\mathcal{D} + (1-\delta)$ is strictly increasing in z in every state, and the issuance multiplier $1 + \xi(1 - 1/q(z))$ is nondecreasing in z (equal to one below z_q, increasing above). So $\mathcal{R}^P(z, \varepsilon)$ rises with z state by state, and since the certainty equivalent respects statewise dominance, $P(\cdot)$ is strictly increasing.

Thus, occupational indifference implies $P(\bar{z}) = H$, and $P(\bar{z}) = H \geq P(z_q)$, and strict monotonicity of P indicates $\bar{z} \geq z_q$. Every producing type $z \geq \bar{z} \geq z_q$ then has $q(z) \geq q(z_q) = 1$ and issues at the cap. □

B.6 Proof of Proposition 3.5

Proof.
Fix a realization of ε. Each active producer $i \in \mathcal{P}$ operates $y_i = (\tilde{z}_i k_i)^\alpha l_i^{1-\alpha}$ and hires labor at the common wage w, so his labor demand implies a common effective capital-labor ratio:

$$y_i = \Big(\frac{\tilde{z}_i k_i}{l_i}\Big)^\alpha l_i, \qquad \frac{\tilde{z}_i k_i}{l_i} = \left(\frac{w}{1-\alpha}\right)^{1/\alpha} \quad \text{for every } i \in \mathcal{P}, \tag{B.23}$$

the same for each i because all face the same w. A ratio that is common across producers equals the ratio of the aggregates,

$$\frac{\tilde{z}_i k_i}{l_i} = \frac{\int_{i\in\mathcal{P}} \tilde{z}_i k_i \, di}{\int_{i\in\mathcal{P}} l_i \, di} = \frac{\exp(\varepsilon)\int_{i\in\mathcal{P}} z_i k_i \, di}{L}, \qquad L \equiv \int_{i\in\mathcal{P}} l_i \, di. \tag{B.24}$$

Summing firm-level output across producers and substituting,

$$Y = \int_{i\in\mathcal{P}} y_i \, di = \left(\frac{\exp(\varepsilon)\int_{i\in\mathcal{P}} z_i k_i \, di}{L}\right)^{\alpha} \int_{i\in\mathcal{P}} l_i \, di = \Big(\exp(\varepsilon)\int_{i\in\mathcal{P}} z_i k_i \, di\Big)^{\alpha} L^{1-\alpha}. \tag{B.25}$$

Capital is determined by the net wealth, $k_i = (1+\xi)a_i$ with a_i independent of z_i, and whether being a producer depends only on z_i. The capital-weighted average of z among producers therefore equals the unweighted truncated mean:

$$\int_{i\in\mathcal{P}} z_i k_i \, di = \mathbb{E}[z \mid z \geq \bar{z}] \int_{i\in\mathcal{P}} k_i \, di = \mathrm{TFP} \times K. \tag{B.26}$$

Substituting,

$$Y = \exp(\alpha\varepsilon)\big(\mathbb{E}[z \mid z \geq \bar{z}]\big)^{\alpha} K^{\alpha} L^{1-\alpha} = Z\, K^{\alpha} L^{1-\alpha}, \qquad Z = \exp(\alpha\varepsilon)\,\mathrm{TFP}^{\alpha}, \tag{B.27}$$

and labor-market clearing sets $L = 1$. Finally, the labor first-order condition gives $w\, l_i = (1-\alpha)\, y_i$ firm by firm; summing across producers, $w\, L = (1-\alpha)\, Y$, and aggregate operating profit is $\Pi^A = \int_{i\in\mathcal{P}} (y_i - w\, l_i)\, di = \alpha\, Y$. □

B.7 Aggregate consumption and investment

Aggregate consumption and investment follow from the flows of the period; primes denote next-period values. Capitalists' aggregate cash on hand is the total payoff on capital claims plus the bond repayment,

$$M = (1-\tau)\,\alpha Y + (1-\delta)K + rB, \tag{B.28}$$

of which they save next period's wealth W' and consume the rest, $C^K = M - W'$. The common saving rate splits this consumption by occupation, $C^K = C^P + C^I$ with $C^P = (1-\varsigma)M^P$ and $C^I = (1-\varsigma)M^I$, where $M^P = W\int_{\bar{z}}^{\infty} \mathcal{R}^P(z,\varepsilon)\, d\Phi(z)$ and $M^I = \Phi(\bar{z})\, \mathcal{R}^I(\varepsilon)\, W$ are the occupations' cash on hand, using that wealth is independent of the draw. Workers consume their wage, the fiscal residual on the maturing bond, and the proceeds of the new issue, which the government transfers to them at the period's settlement:

$$C^W = \underbrace{(1-\alpha)Y}_{\text{wage}} + \underbrace{\tau\alpha Y - rB}_{\text{fiscal residual}} + \underbrace{B'}_{\text{new-issue proceeds}}. \tag{B.29}$$

Aggregate investment is the change in the capital stock net of depreciation, which can be written as

$$I = K' - (1-\delta)K = (1+\xi)\,\bar{z}'^{-\eta}\,W' - (1-\delta)K : \tag{B.30}$$

new capital is the part of capitalist saving that flows to next period's producers, levered by outside equity, while the undepreciated stock carries over. In a stationary equilibrium $K' = K$ and investment is pure replacement, $I = \delta K$. Combining these flows, the goods market clears:

$$Y = \underbrace{C^K + C^W}_{\text{consumption}} + I. \tag{B.31}$$

B.8 Proof of Proposition 4.1

Proof.

By Proposition 3.2, the marginal utility of end-of-period cash on hand carries the continuation weight, so the kernels below are the weighted kernels of Section 4. Consider the foreign investor at the corner portfolio, $\theta_H^F = 0$, with the bond and own-equity holdings interior, so his kernel $\mathcal{M}_F^0$ satisfies $\mathbb{E}[\mathcal{M}_F^0] = 1/r$ and $\mathbb{E}[\mathcal{M}_F^0 \mathcal{R}_F^e] = 1$. The portfolio problem is concave, so zero holdings of Home equity are optimal if and only if the derivative of the objective in θ_H^F is non-positive at the corner:

$$\mathbb{E}\big[\Lambda_{F,t}\, U'(\mathcal{R}_F^I a')\,(\mathcal{R}_H^e - \chi_H - r)\big] \le 0 \quad \Longleftrightarrow \quad \mathbb{E}\big[\mathcal{M}_F^0\,(\mathcal{R}_H^e - \chi_H - r)\big] \le 0, \tag{B.32}$$

rescaling marginal utility by its positive normalization. Since χ_H and r are deterministic, $\mathbb{E}[\mathcal{M}_F^0\,(\chi_H + r)] = (\chi_H + r)/r$, and the condition rearranges to

$$\chi_H \;\ge\; \bar{\chi}_H \equiv r\,\big(\mathbb{E}[\mathcal{M}_F^0\,\mathcal{R}_H^e] - 1\big). \tag{B.33}$$

At the corner, $\mathcal{R}_F^I = r + \theta_F^F(\mathcal{R}_F^e - r)$ is bounded away from zero, because the index return inherits the positive floor of the safe component $(1-\delta)/r$ of q_F, so marginal utility is bounded and $\mathbb{E}[\mathcal{M}_F^0\,\mathcal{R}_H^e]$ is a finite number. Expanding the expectation with $\mathbb{E}[\mathcal{M}_F^0] = 1/r$ gives the premium-comovement decomposition

$$\bar{\chi}_H = \big(\mathbb{E}[\mathcal{R}_H^e] - r\big) + r\ \mathrm{Cov}\,\big(\mathcal{M}_F^0, \mathcal{R}_H^e\big), \tag{B.34}$$

and subtracting the Home investor's own-equity condition $\mathbb{E}[\mathcal{M}_H^0 \mathcal{R}_H^e] = 1$, which holds with equality at the same corner equilibrium because Home holdings of Home equity remain interior there, gives the covariance comparison

$$\bar{\chi}_H = r\,\big[\,\mathrm{Cov}(\mathcal{M}_F^0, \mathcal{R}_H^e) - \mathrm{Cov}(\mathcal{M}_H^0, \mathcal{R}_H^e)\big], \tag{B.35}$$

using $\mathbb{E}[\mathcal{M}_F^0] = \mathbb{E}[\mathcal{M}_H^0] = 1/r$. Foreign investors hold no Home equity when the investment wedge exceeds the risk-adjusted benefit of adding it to their portfolio. This benefit is evaluated at zero Home-equity holdings, and the threshold depends on equilibrium returns and the investor's valuation of risk. □

C Additional Model Results

This appendix presents the full comparative statics panels for each parameter exercise discussed in Section 5. Each panel reports capital variables, financial variables, and macroeconomic aggregates as functions of the parameter being varied. In the financial panels, the expected portfolio return r^p is the expected return on the portfolio of Home investors, with Foreign's drawn in dashed lines; and the portfolio shares are drawn as solid lines for Home investors and dashed lines for Foreign investors. In the macroeconomic panels the wage satisfies $w = (1-\alpha)Y$ state by state.

Figures C.1–C.3 report the full set of equilibrium variables as the home tax parameter τ is varied from its symmetric benchmark. Figures C.4–C.6 report the results as home initial wealth A is varied. Figures C.7–C.9 report the results as the Pareto shape parameter η for the home country is varied. Figures C.10–C.12 report results as the home equity financing constraint ξ is varied.

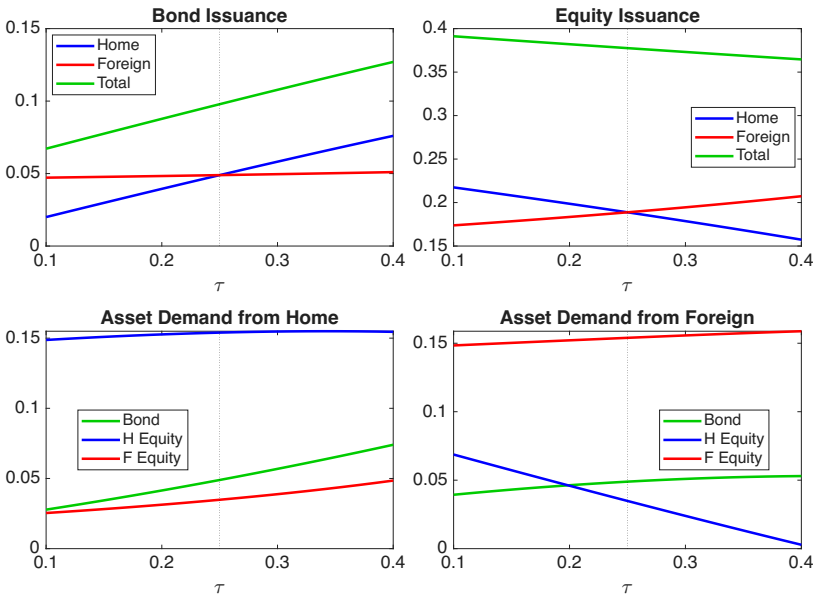


Figure C.1: Capital and production variables: varying τ.

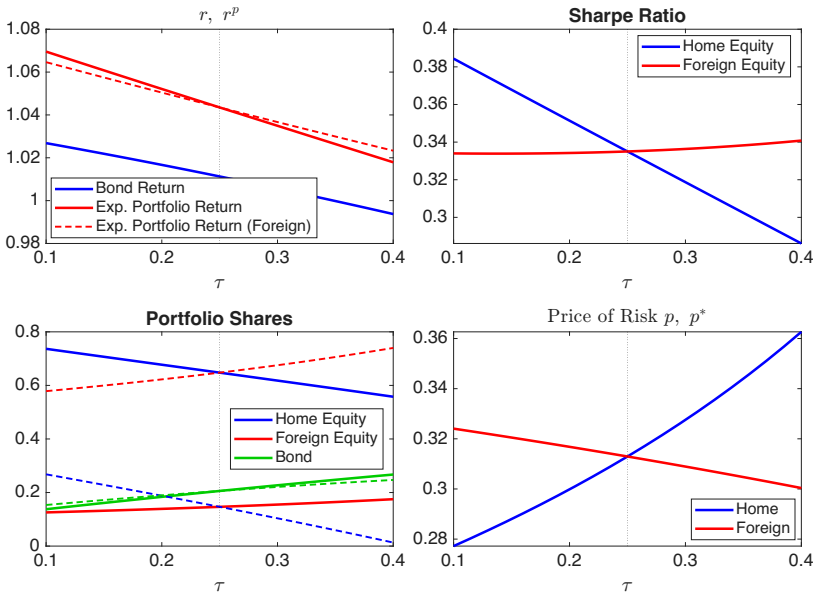


Figure C.2: Financial variables: varying τ.

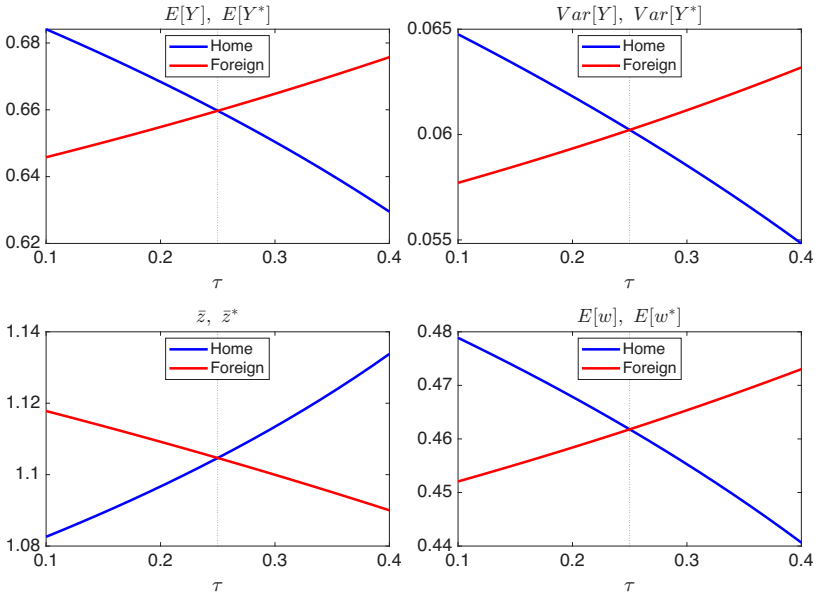


Figure C.3: Macroeconomic aggregates: varying τ.

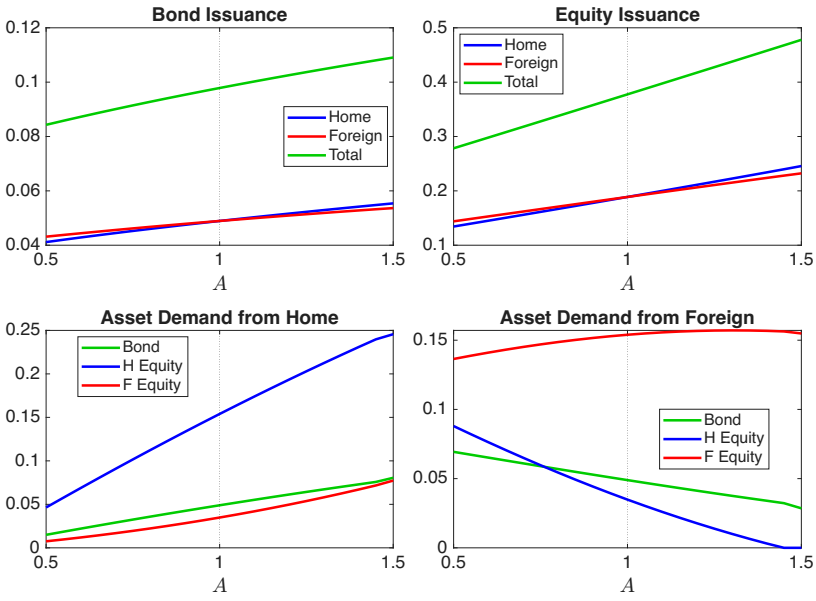


Figure C.4: Capital and production variables: varying A.

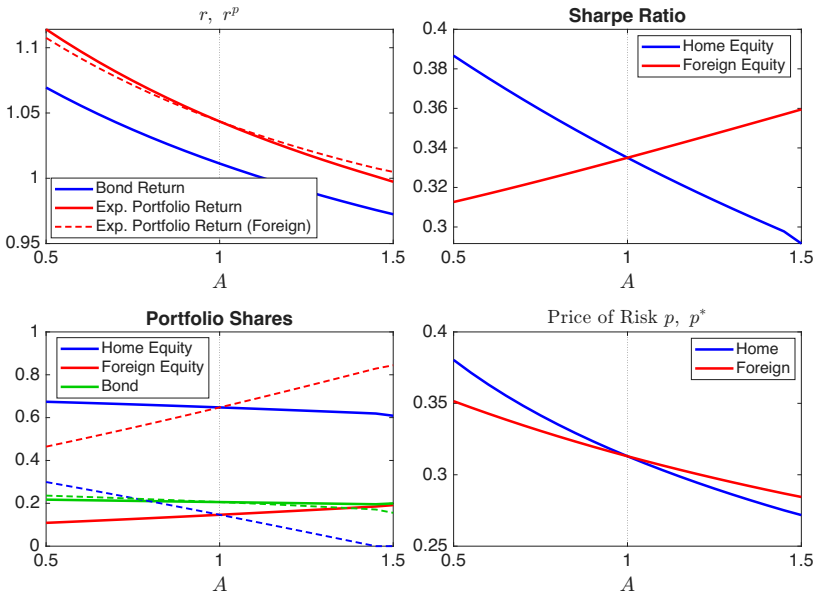


Figure C.5: Financial variables: varying A.

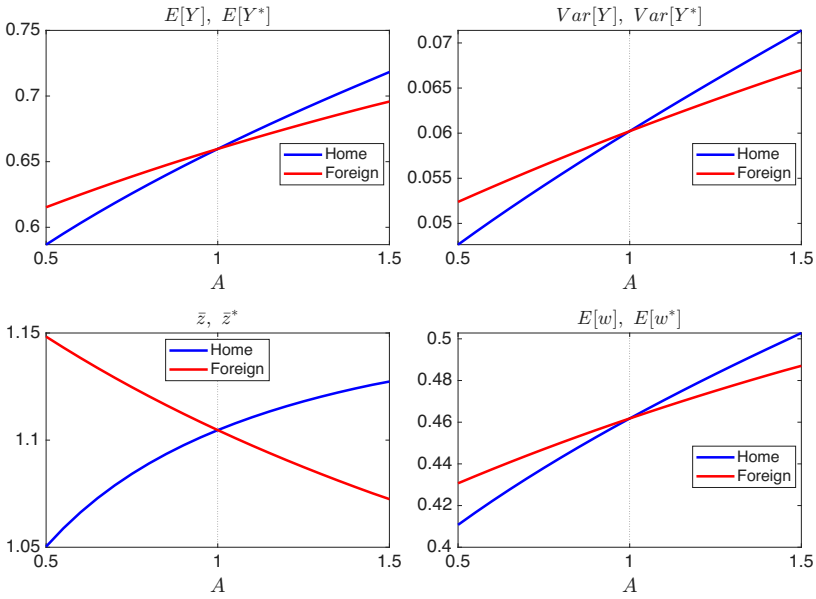


Figure C.6: Macroeconomic aggregates: varying A.

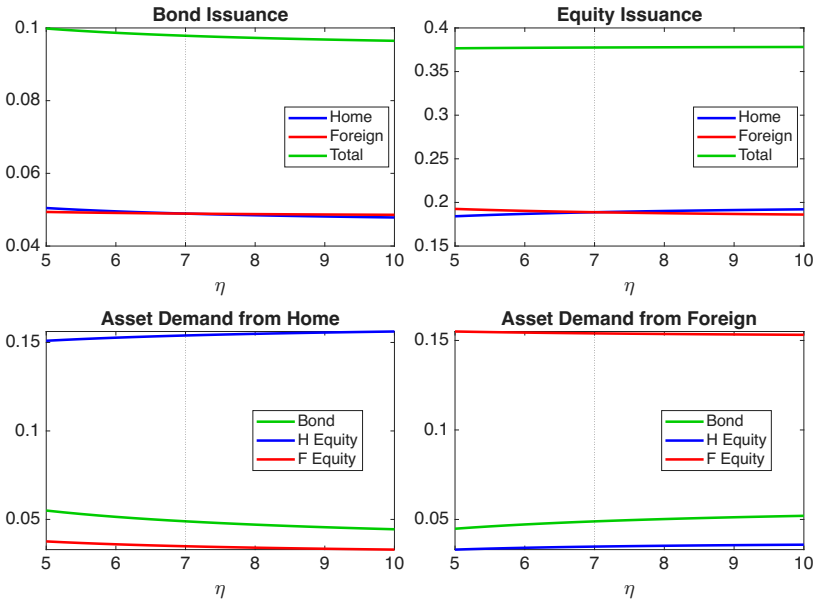


Figure C.7: Capital and production variables: varying η.

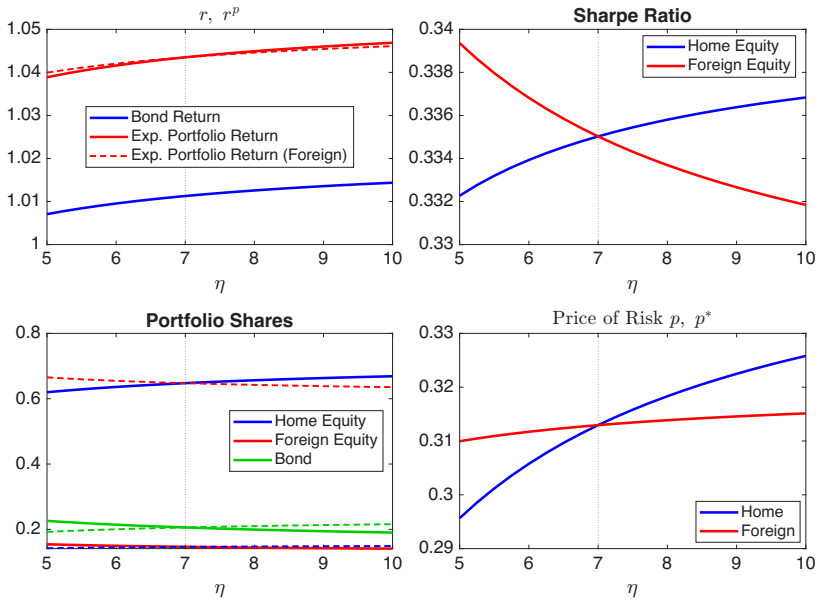


Figure C.8: Financial variables: varying η.

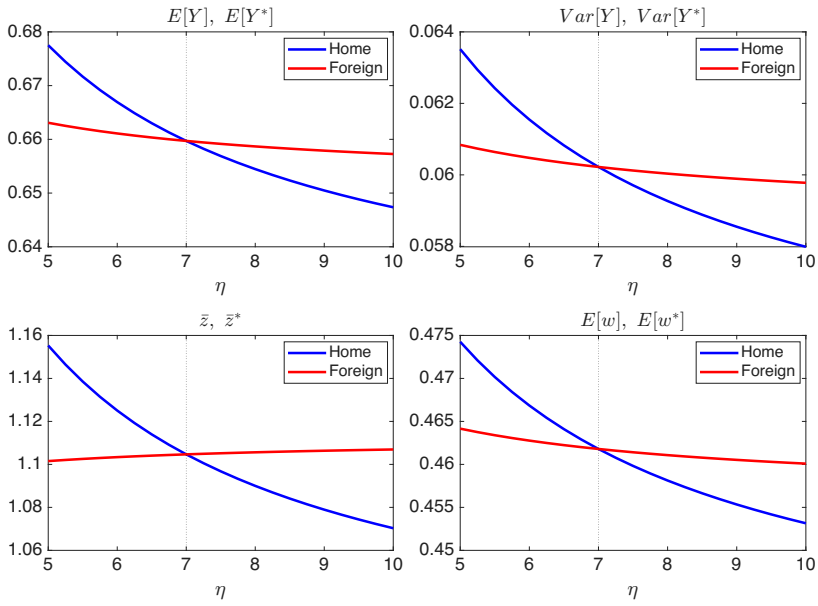


Figure C.9: Macroeconomic aggregates: varying η.

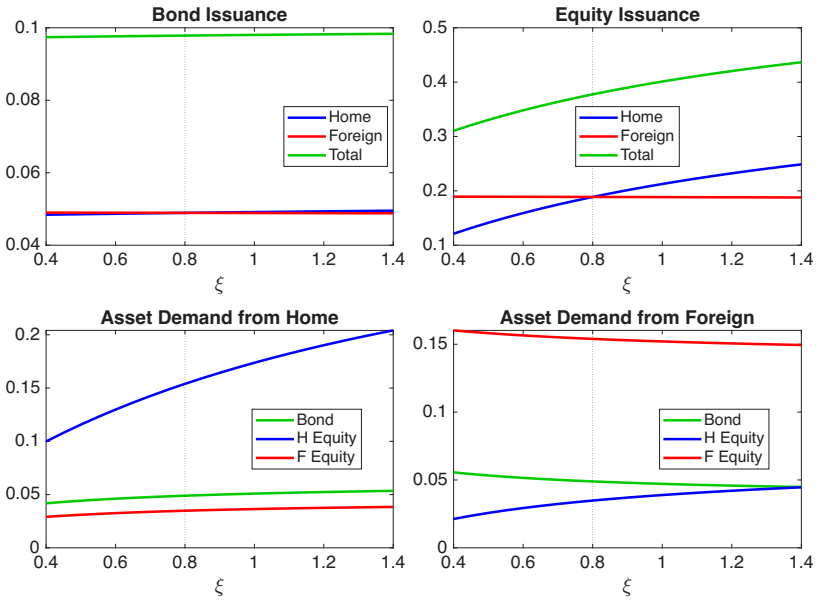


Figure C.10: Capital and production variables: varying ξ.

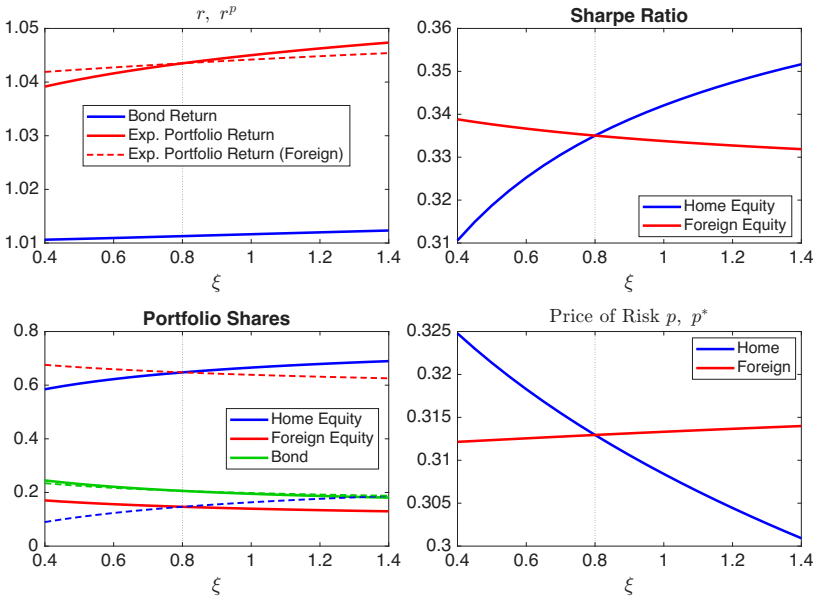


Figure C.11: Financial variables: varying ξ.

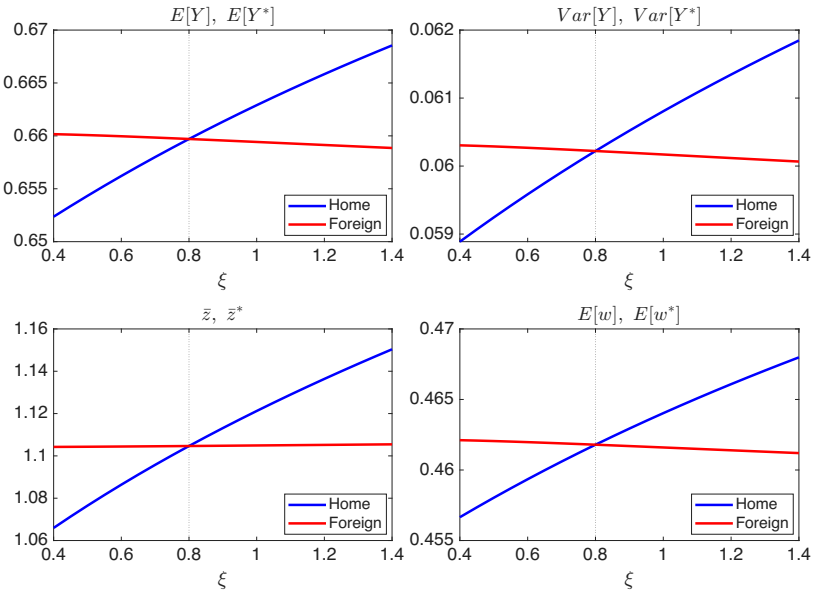


Figure C.12: Macroeconomic aggregates: varying ξ.